

BABU BANARSI DAS UNIVERSITY

LUCKNOW

SESSION: 2024-25



SCHOOL OF COMPUTER APPLICATION

Project on

DESCRIPTIVE ANALYTICS (BCADSN13201)

Submitted by

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Submitted to

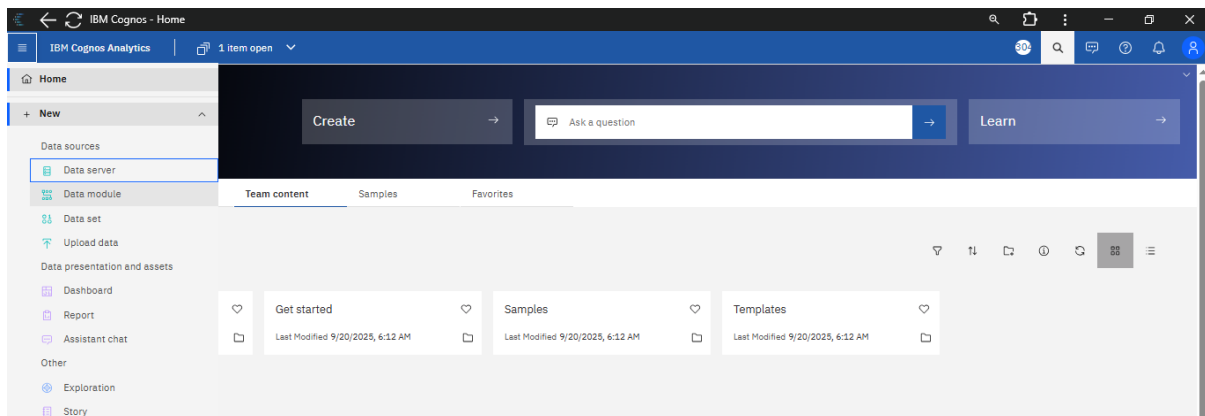
Mr. Ayush Chand

Project

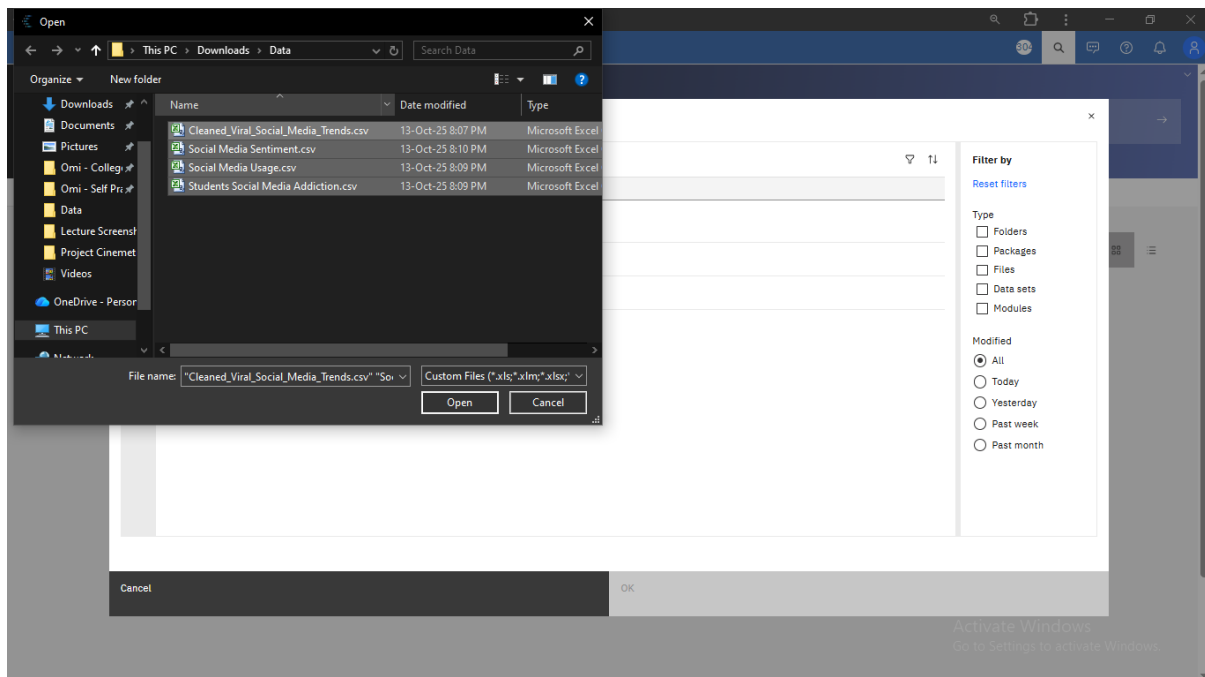
Problem Statement: Analyse the average screen time per platform across user

Solution : Create a Bar Chart using Data Module

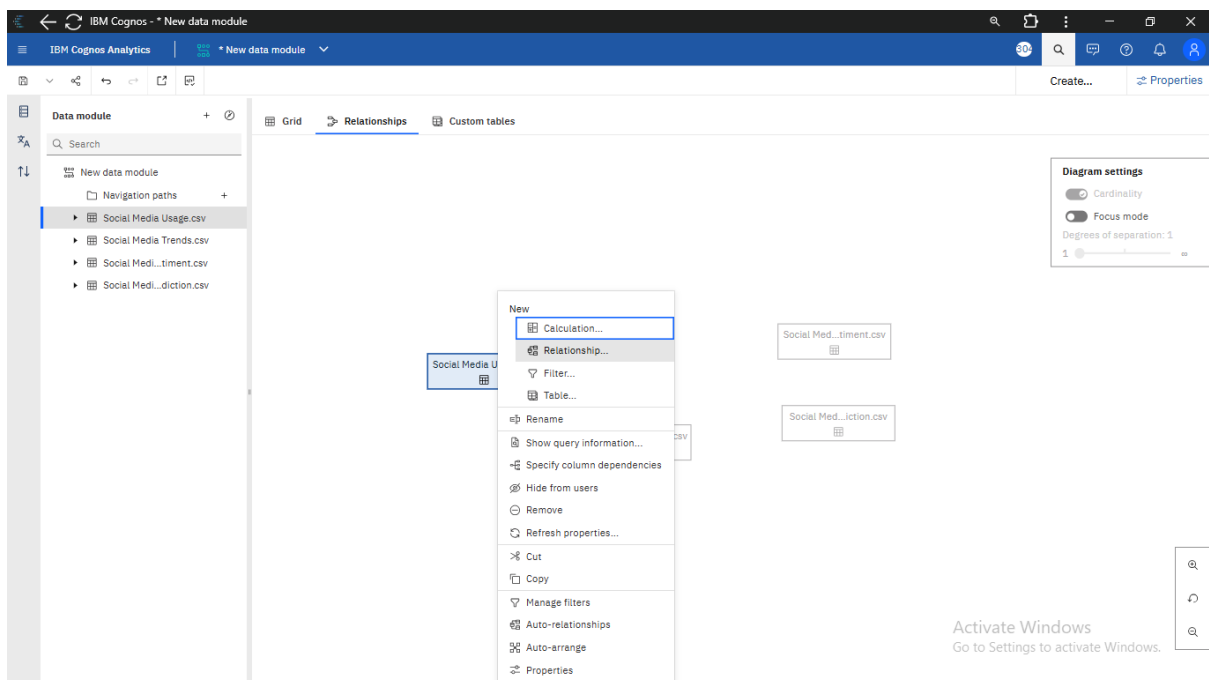
Step 1: Login to IBM Cognos and following interface will be opened. Click Three-Line Icon and Select Data Module option.



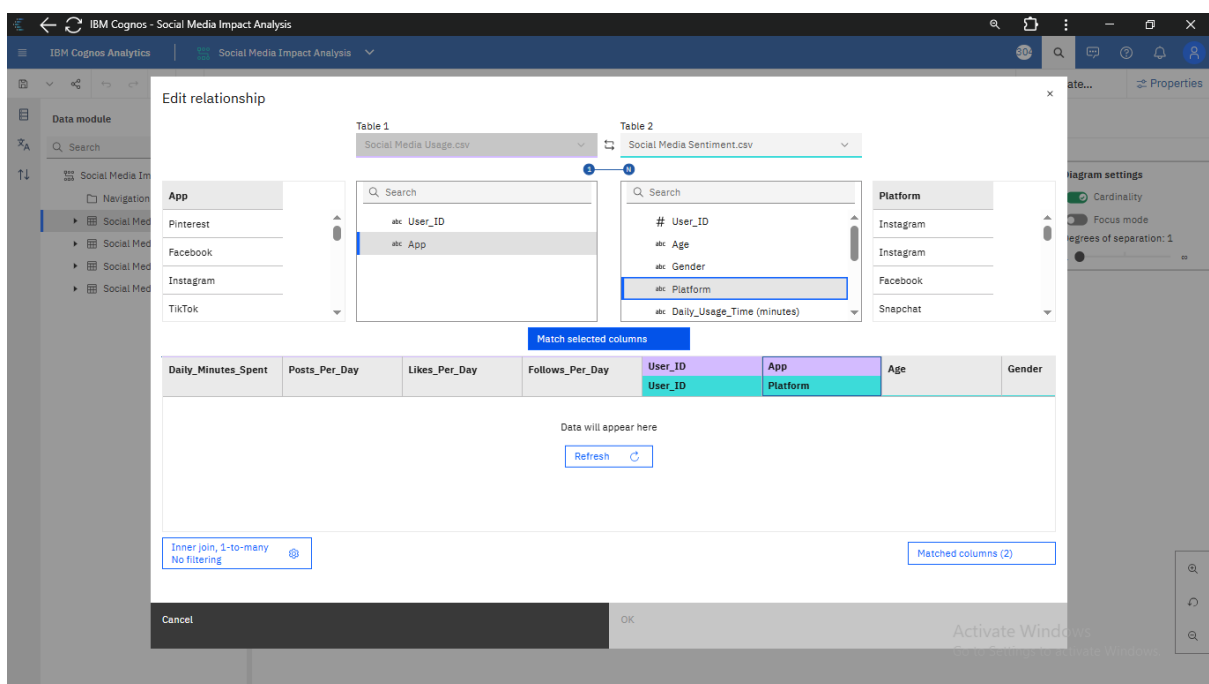
Step 2: Click on upload icon and Select all the files you want to use.



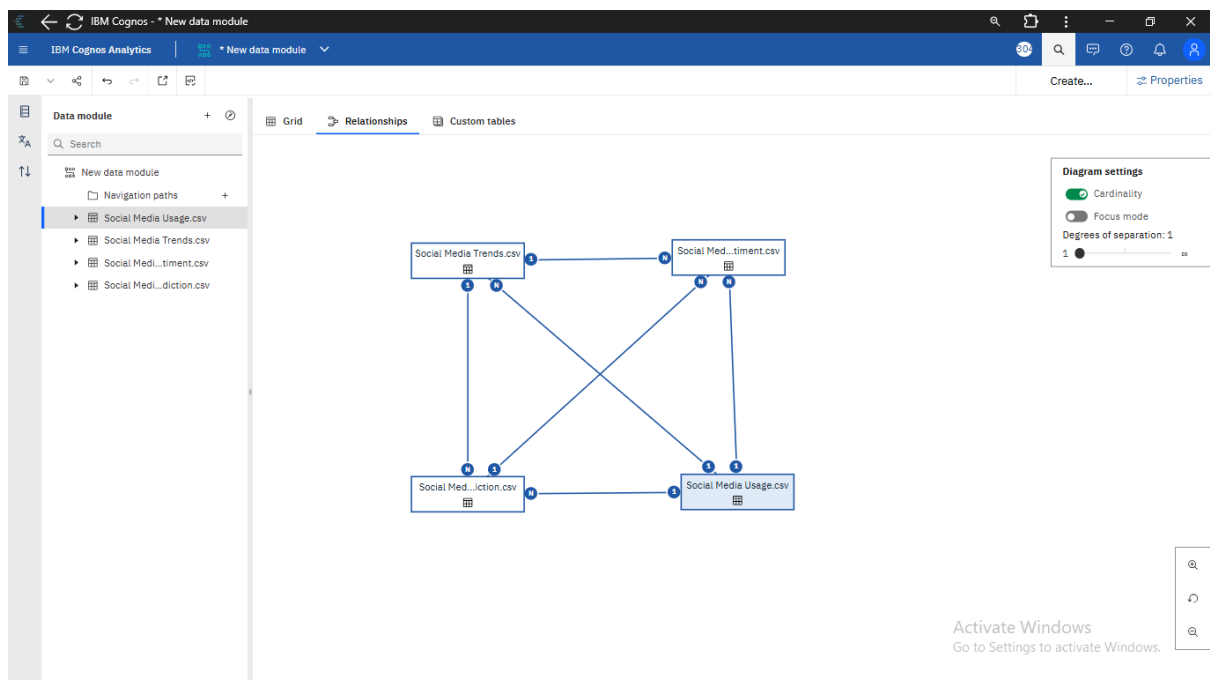
Step 3: Click on the Relationship Option on the canvas, Select the desired dataset and left-click and select Relationships from given options



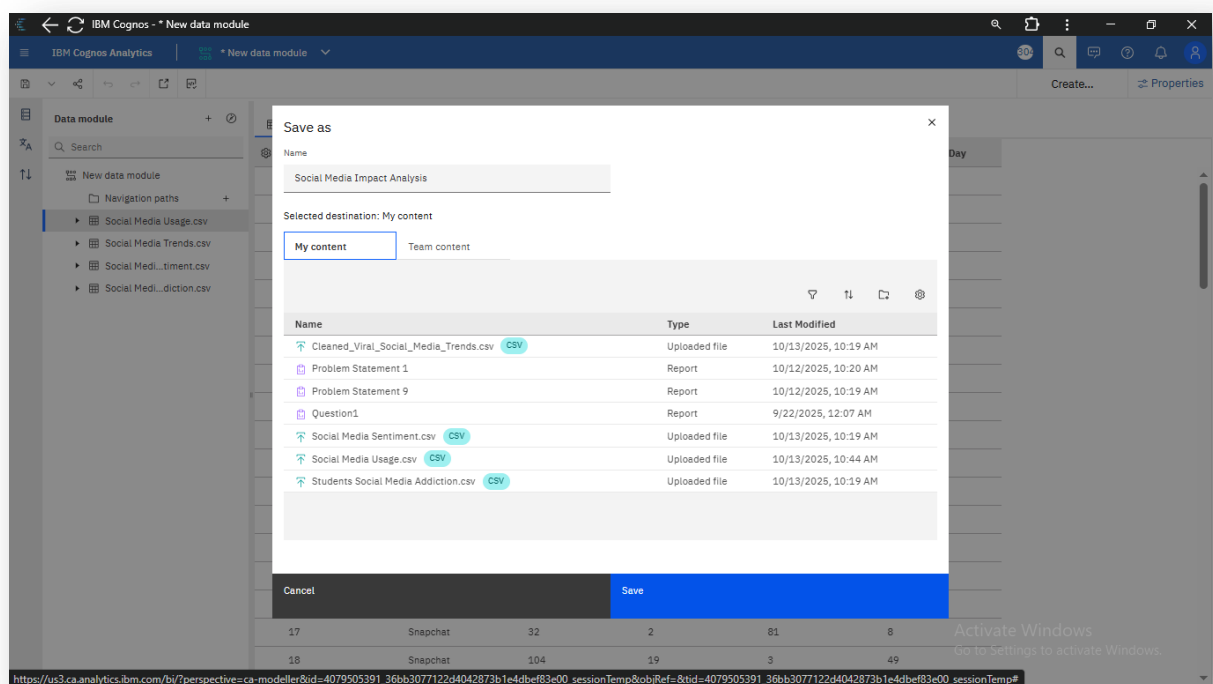
Step 4: Select Table-2 from drop-down option and click on match selected columns and Do this all the datasets



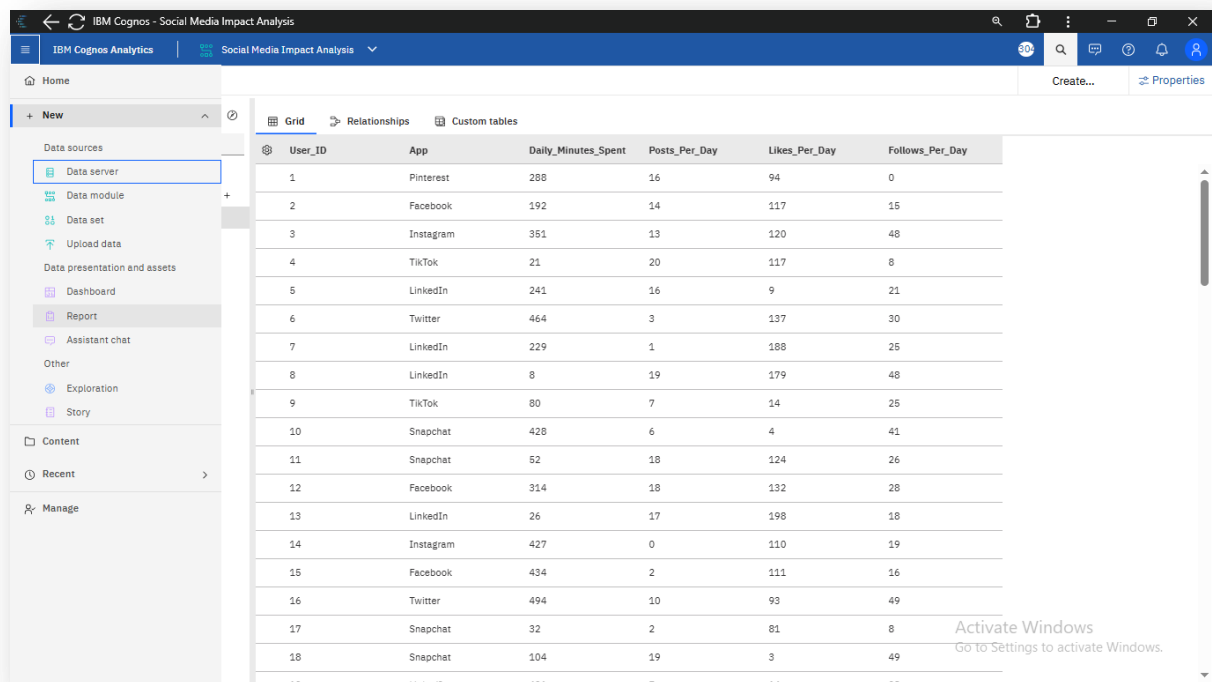
Step 5: Establish Relationship between Datasets and following visuals will appear



Step 6: Now Click on save option and Give Module an appropriate name, Select My content and click on Save



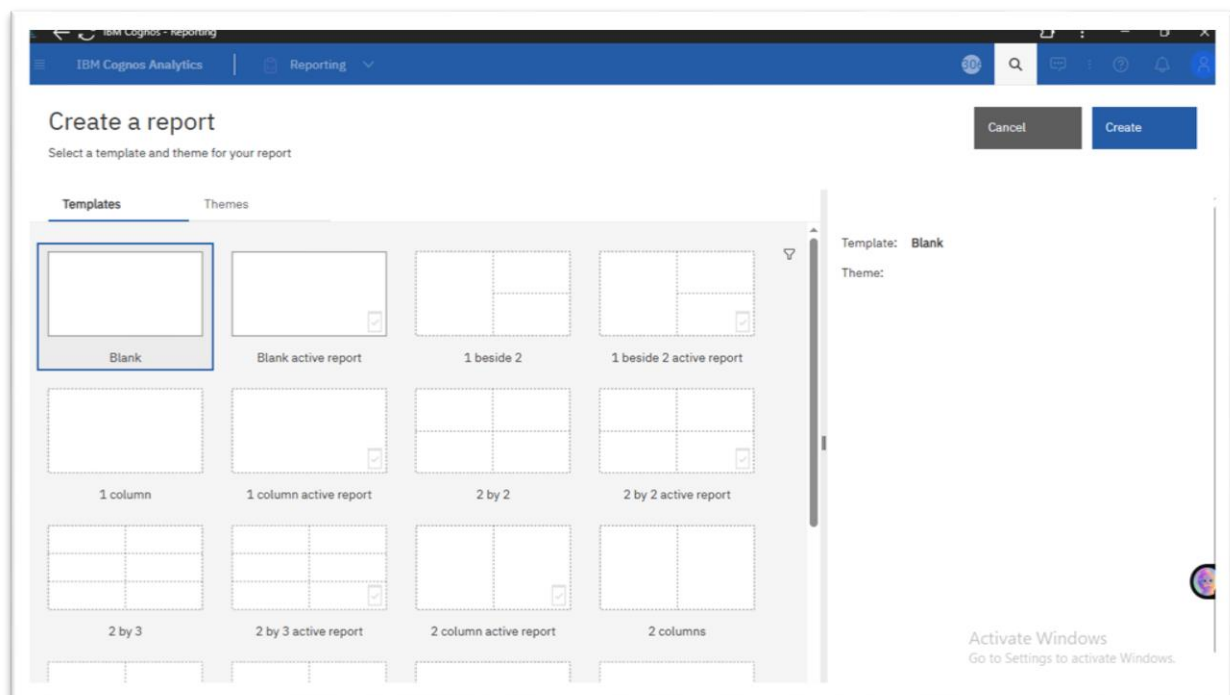
Step 7: Click on Three-Line icon again and select New-> Report option



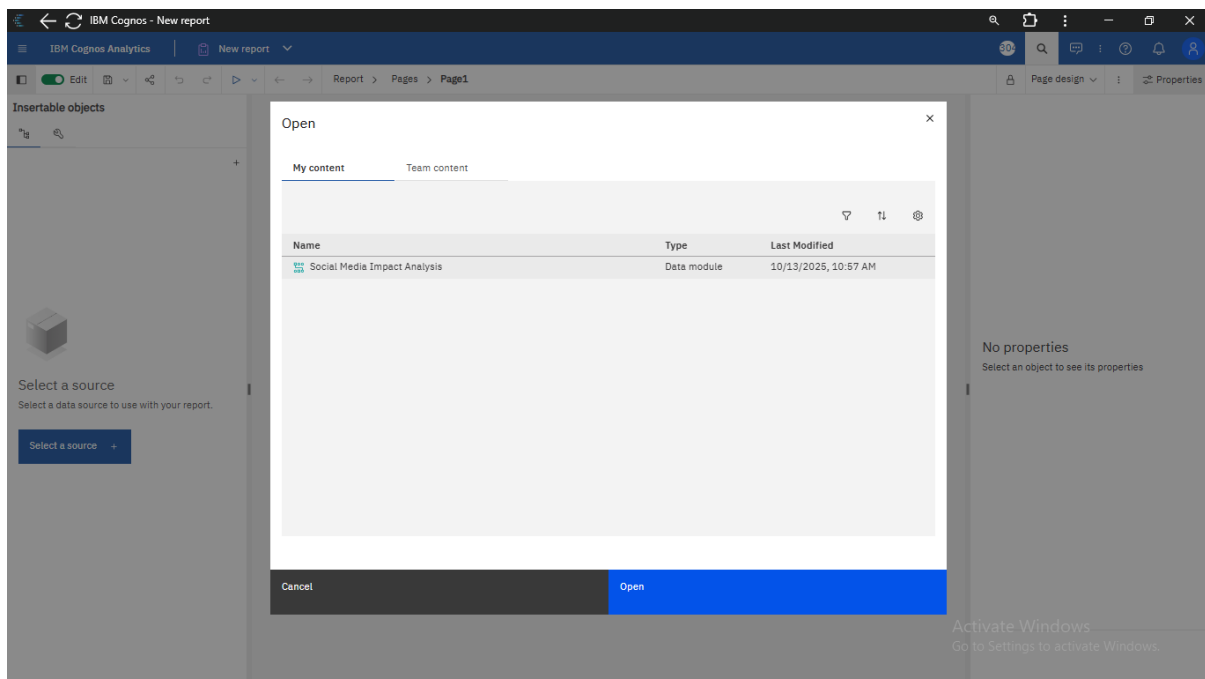
The screenshot shows the IBM Cognos Analytics interface for 'Social Media Impact Analysis'. The left sidebar has a 'New' button at the top, and below it, a 'Data sources' section with 'Data server' selected. The main area displays a grid table with the following data:

User_ID	App	Daily_Minutes_Spent	Posts_Per_Day	Likes_Per_Day	Follows_Per_Day
1	Pinterest	288	16	94	0
2	Facebook	192	14	117	15
3	Instagram	351	13	120	48
4	TikTok	21	20	117	8
5	LinkedIn	241	16	9	21
6	Twitter	464	3	137	30
7	LinkedIn	229	1	188	25
8	LinkedIn	8	19	179	48
9	TikTok	80	7	14	25
10	Snapchat	428	6	4	41
11	Snapchat	52	18	124	26
12	Facebook	314	18	132	28
13	LinkedIn	26	17	198	18
14	Instagram	427	0	110	19
15	Facebook	434	2	111	16
16	Twitter	494	10	93	49
17	Snapchat	32	2	81	8
18	Snapchat	104	19	3	49

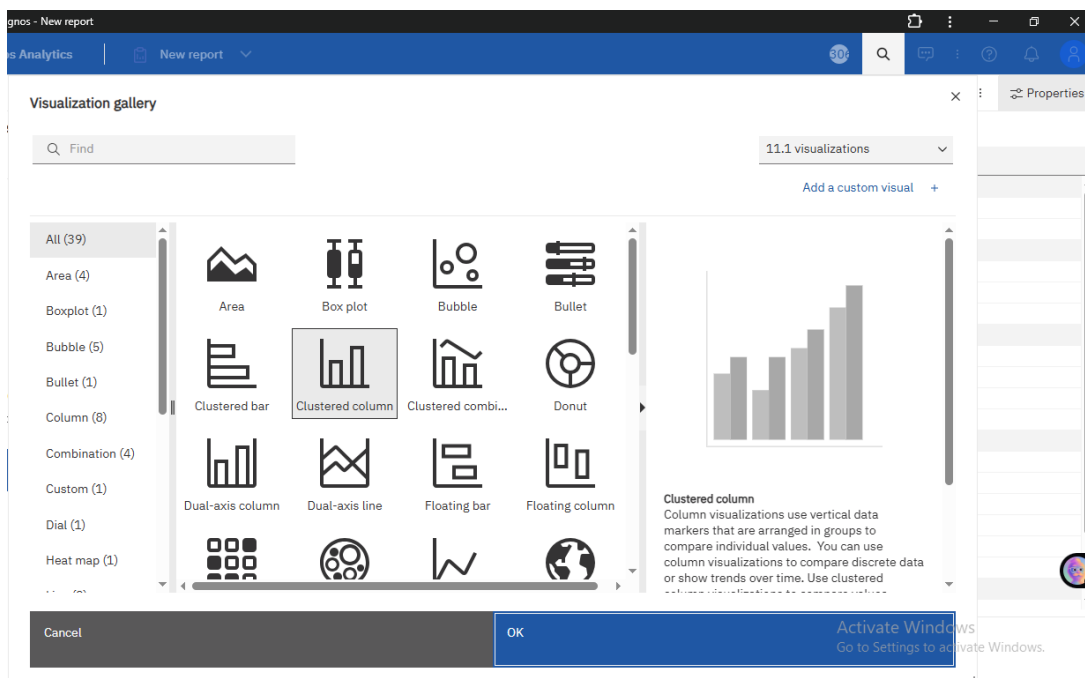
Step 8: Click on Blank Report Option and Then Click on Create Option



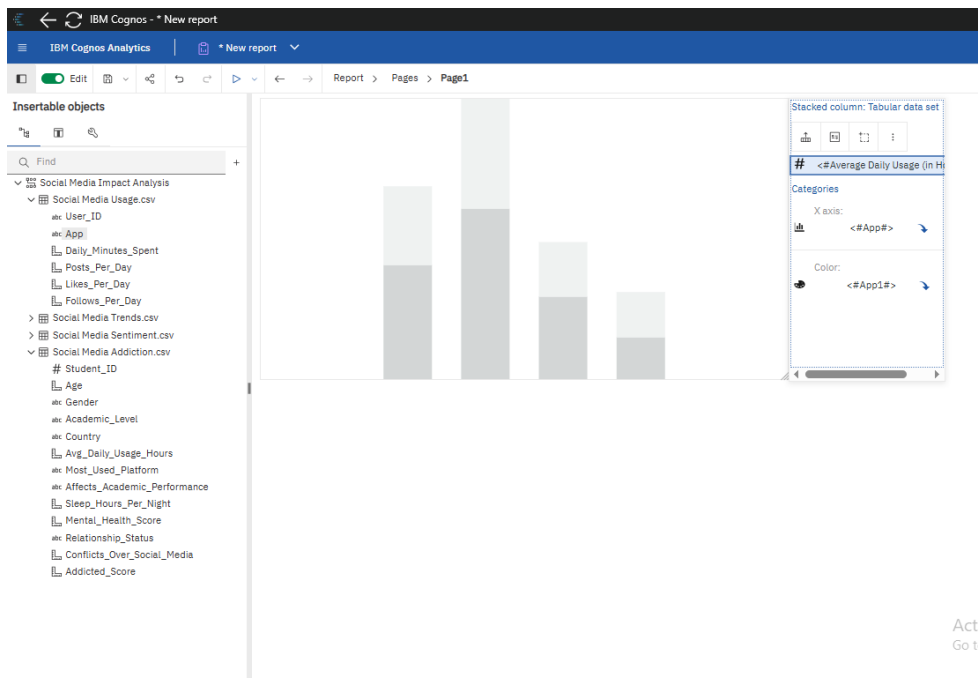
Step 9: Select Data Module from My Content Option and Click on Open.



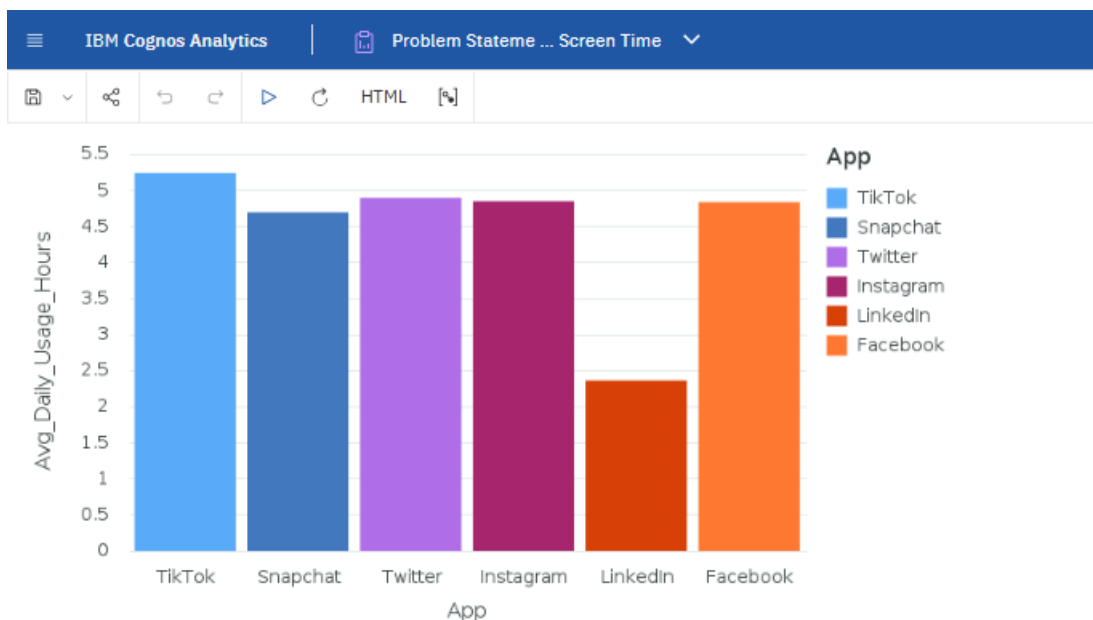
Step 10: Select Visualization Widget from the given options and choose Clustered columns



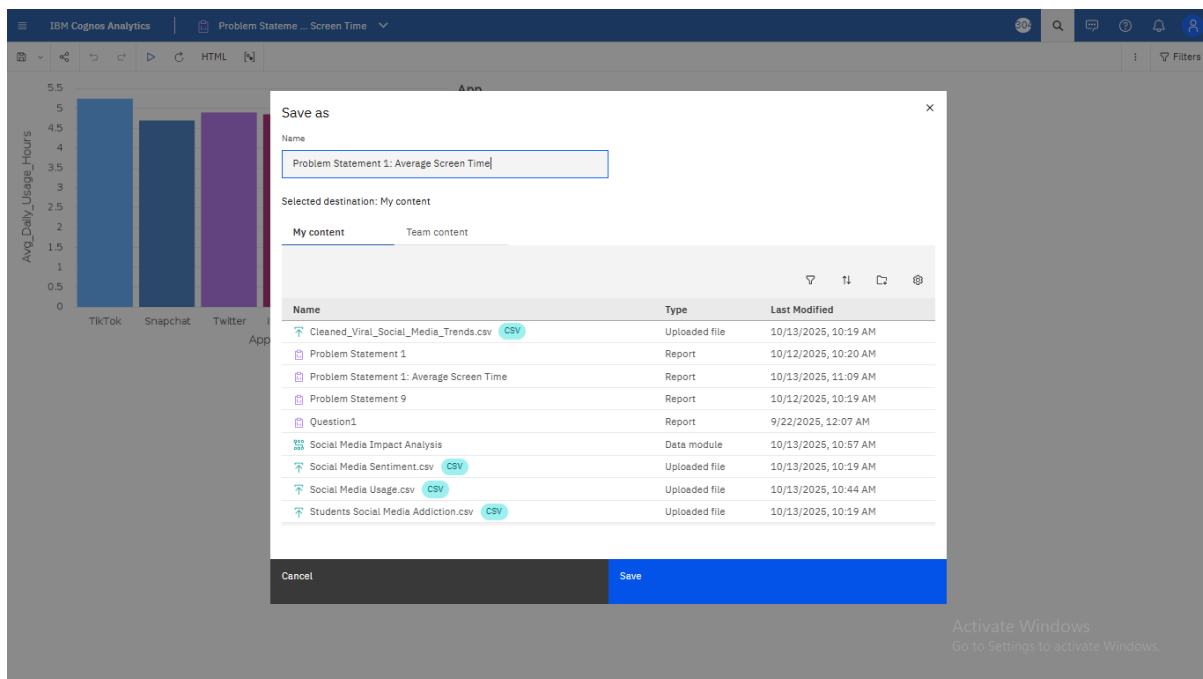
Step 11: Drag 'App' from Social Media Usage into X-axis and Colour, and Average Daily Usage in Y-axis options



Step 12: Run the report and following output will be generated



Step 13: Save the Report with the Desired name.

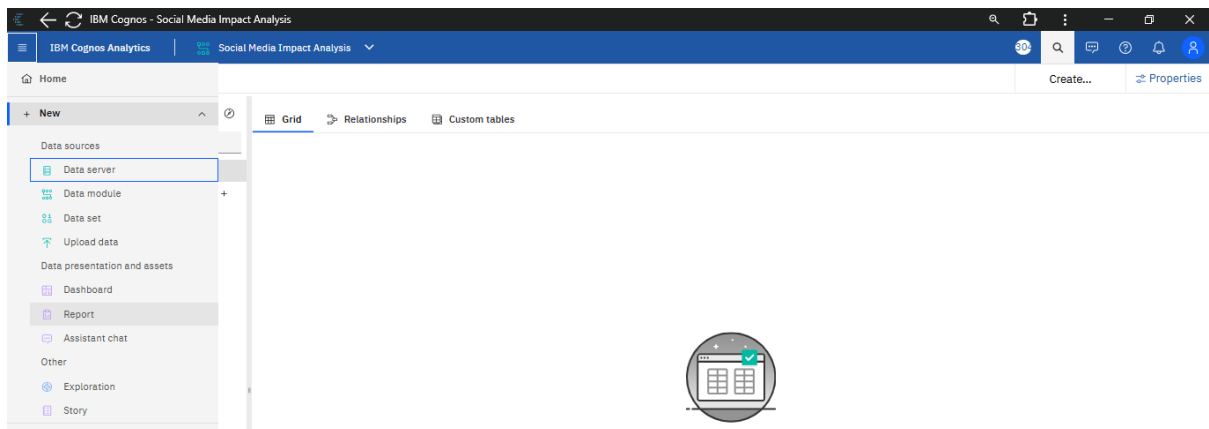


Project

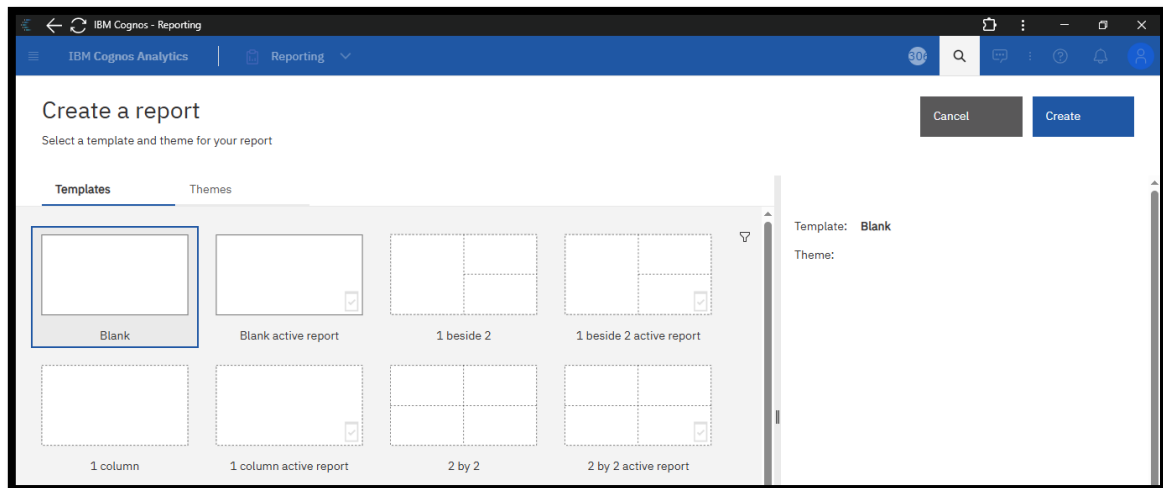
Problem Statement: Rank countries or regions by their average mental health score and usage level

Solution: Create a Map Report using Data Module

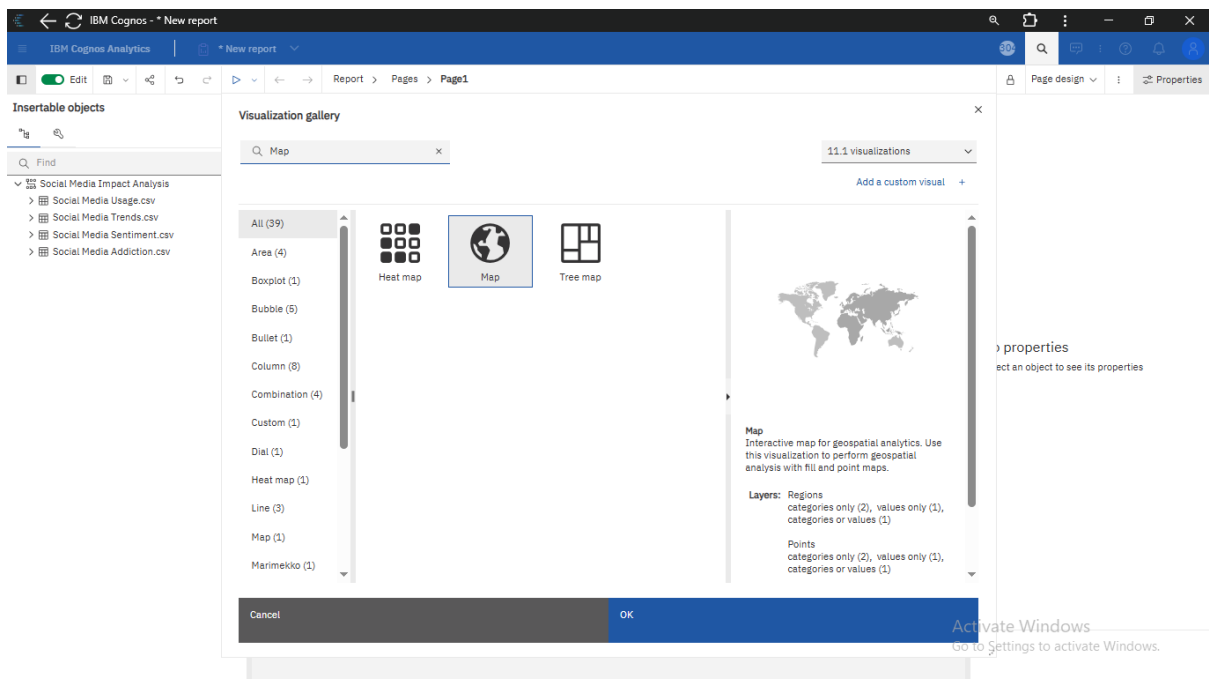
Step 1: Click on Three-Line Icon and Select New -> Report option.



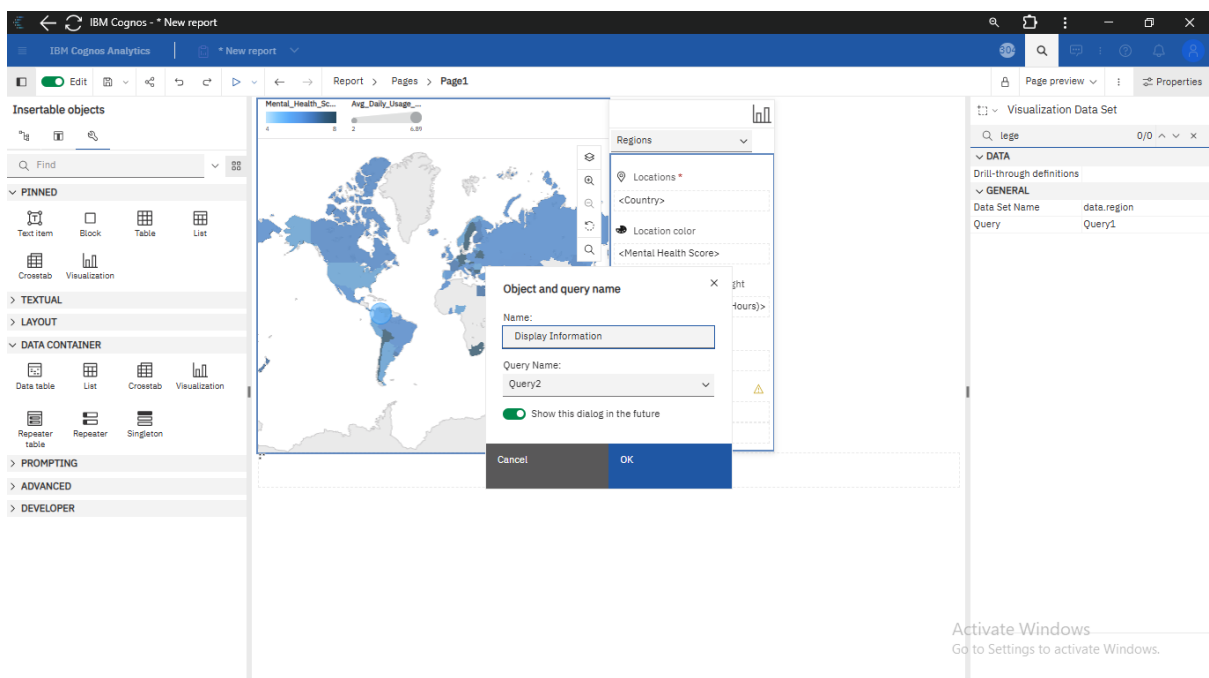
Step 2: Select Blank Report Option and Click on Create



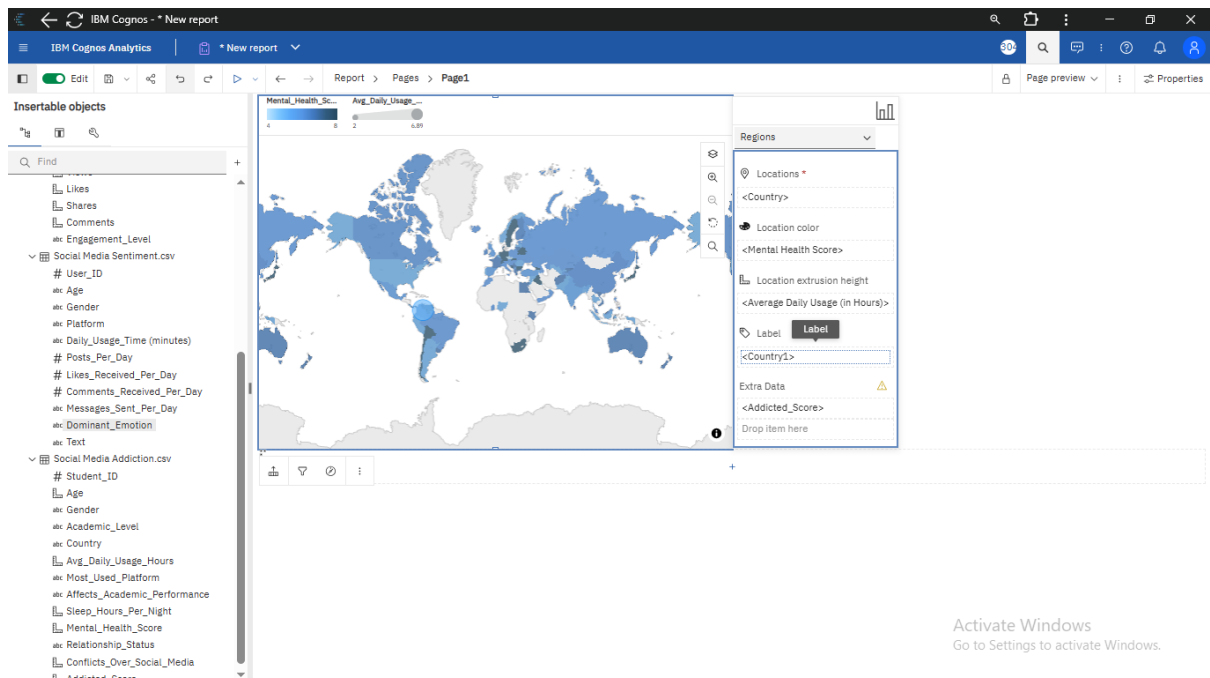
Step 3: Select Visualization Widget and Click on Map option



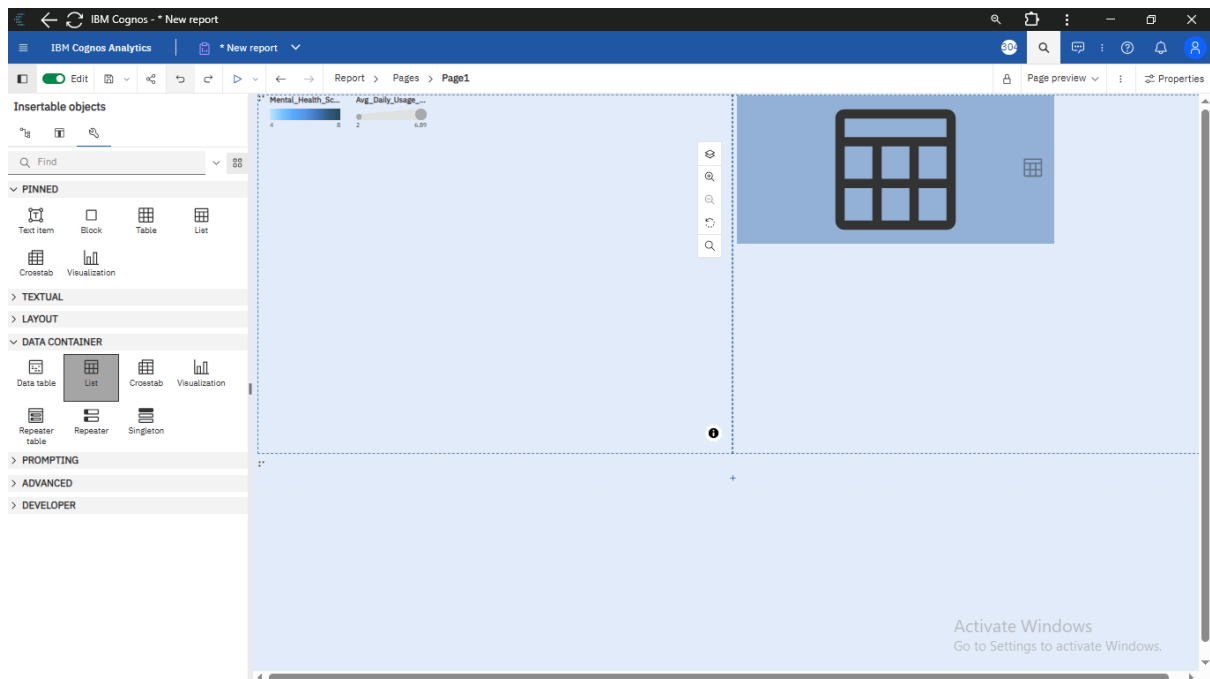
Step 4: Give a suitable name to it.



Step 6: Drag and Drop Country in Location, Mental Health Score in Location colour, Average daily usage in Location height and Country in labels option



Step 7: Now in Click on Tool-icon in Insertable objects and Drag and Drop List from Data container option



Step 8: Select Desired Columns then Click on Conditional Styles in Properties Panel, Create desired conditions by clicking on “+” icon

The screenshot shows the IBM Cognos Analytics interface. On the left, a world map is displayed with a color scale for 'Avg_Daily_Usage_Hours'. In the center, a data table lists countries and their usage hours. A 'Conditional style - advanced' dialog is open, showing a condition '[Query2].[Avg_Daily_Usage_Hours] >= 5.500' with a yellow background color. The Properties panel on the right shows the 'Conditional styles' section.

Country	Avg_Daily_Usage_Hours	Usage_Hours	Usage_Hours	Usage_Hours
Australia	4.6875	6.75	5.5	Low
	4.6875	6.75	5.5	Medium
	4.6875	6.75	5.5	High
Brazil	5.91820986	6	7	Medium
	5.88873239	6	7	High

Conditional style - advanced

Name: High Usage Hour

Advanced condition: [Query2].[Avg_Daily_Usage_Hours] >= 5.500

Style: (Custom) AaBbCc

Remaining values (including future values): (Default) AaBbCc

Cancel OK

Properties Panel:

- Conditional styles
- Style variable
- Text source variable
- TEXT SOURCE
- Source type
- Data item value
- DATA
- Data format
- Drill-through definitions
- Group span
- BOX
- Border
- Padding
- Box type
- COLOR & BACKGROUND
- Background image
- Background effects
- Background color
- Foreground color
- FONT & TEXT
- Font
- Horizontal alignment
- Vertical alignment
- White space
- Spacing & breaking
- Direction & justification
- POSITIONING

Step 9: Select formatting options by clicking on Edit button and Click on OK. Do the same for all the desired columns

The screenshot shows the IBM Cognos Analytics interface. On the left, a world map is displayed with a color scale for 'Avg_Daily_Usage_Hours'. In the center, a data table lists countries and their usage hours. A 'Style' dialog is open, showing formatting options like Background color, Font, Bold, Border, Horizontal alignment, Vertical alignment, Relative alignment, and Data format. The Properties panel on the right shows the 'Conditional styles' section.

Country	Avg_Daily_Usage_Hours	Usage_Hours	Usage_Hours	Usage_Hours
Australia	4.6875	6.75	5.5	Low
	4.6875	6.75	5.5	Medium
	4.6875	6.75	5.5	High
Brazil	5.91820986	6	7	Medium
	5.88873239	6	7	High

Style

Basic Advanced

Name: High Usage

Background color: (Yellow)

Foreground color: (Red)

Advanced condition: [Query2].[Avg_Daily_Usage_Hours] >= 5.500

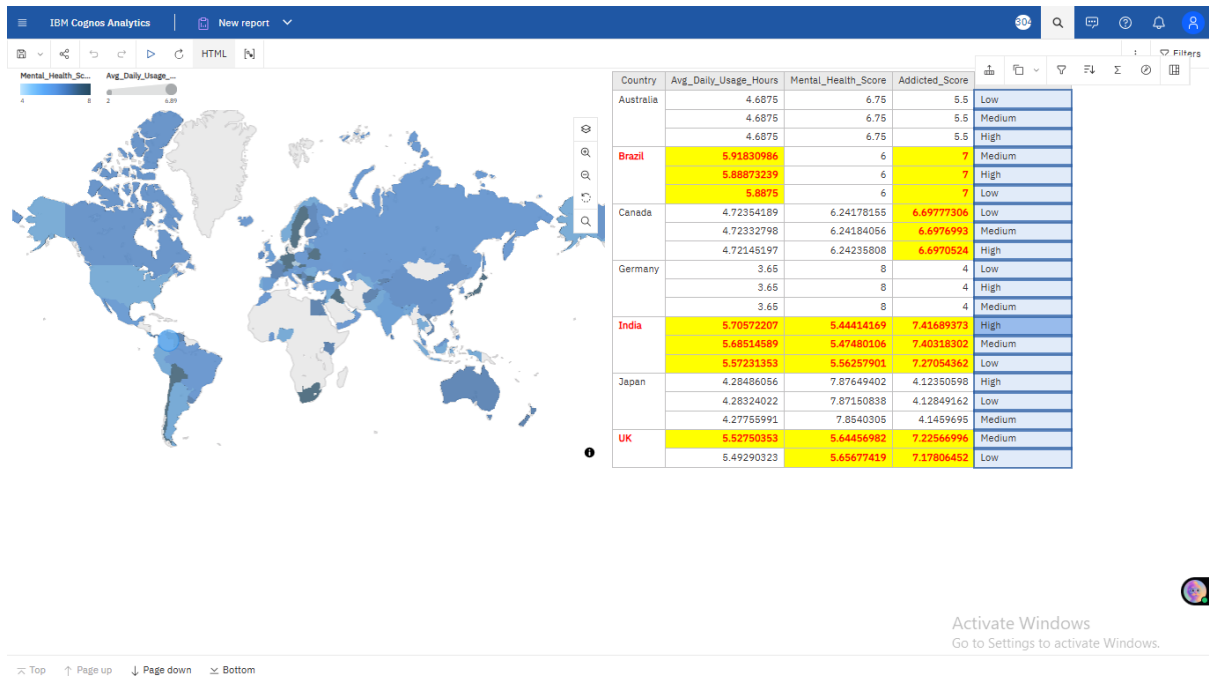
Remaining values (including future values): (Default) AaBbCc

Cancel OK

Properties Panel:

- Conditional styles
- Style variable
- Text source variable
- TEXT SOURCE
- Source type
- Data item value
- DATA
- Data format
- Drill-through definitions
- Group span
- BOX
- Border
- Padding
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- Background effects
- Background color
- Foreground color
- FONT & TEXT
- Font
- Horizontal alignment
- Vertical alignment
- White space
- Spacing & breaking
- Direction & justification
- POSITIONING

Step 10: Run the Report and Save it with suitable name.

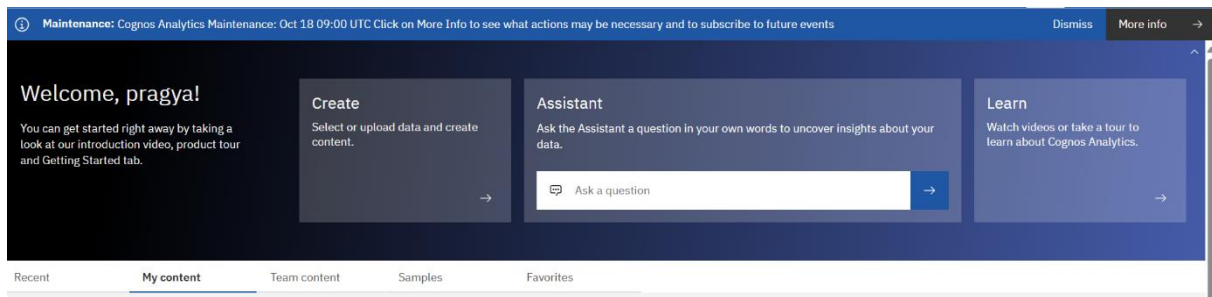


Project

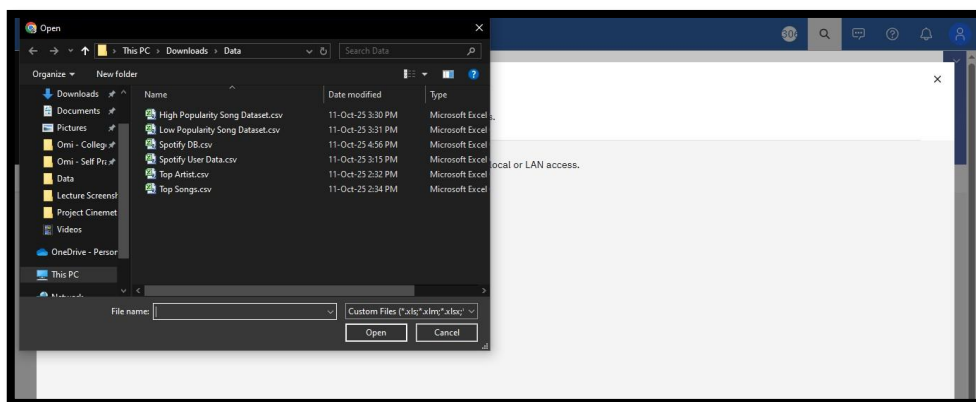
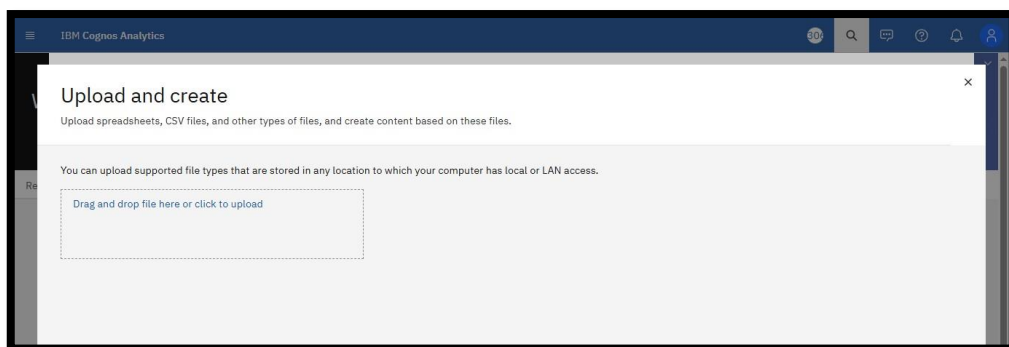
Problem Statement 1: Evaluate gender-wise differences in usage addiction

Solution : Create a Bar Chart using Data set

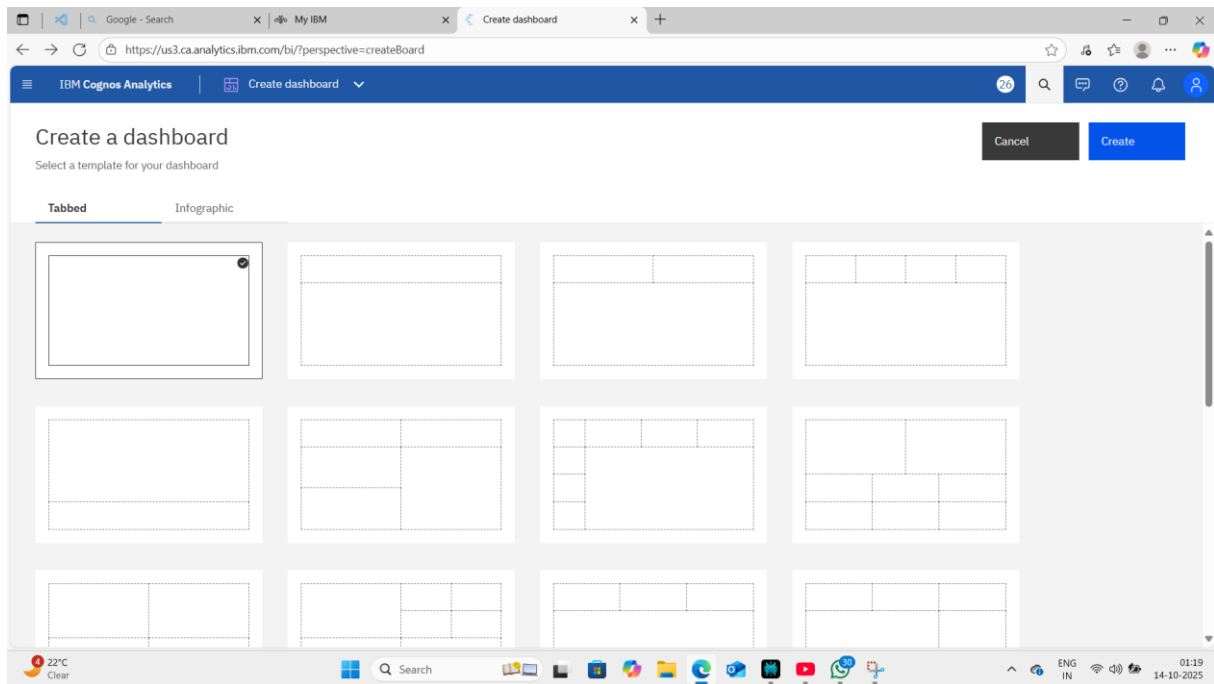
Step 1: Login to IBM Cognos and following interface will be opened. Click on Create option.



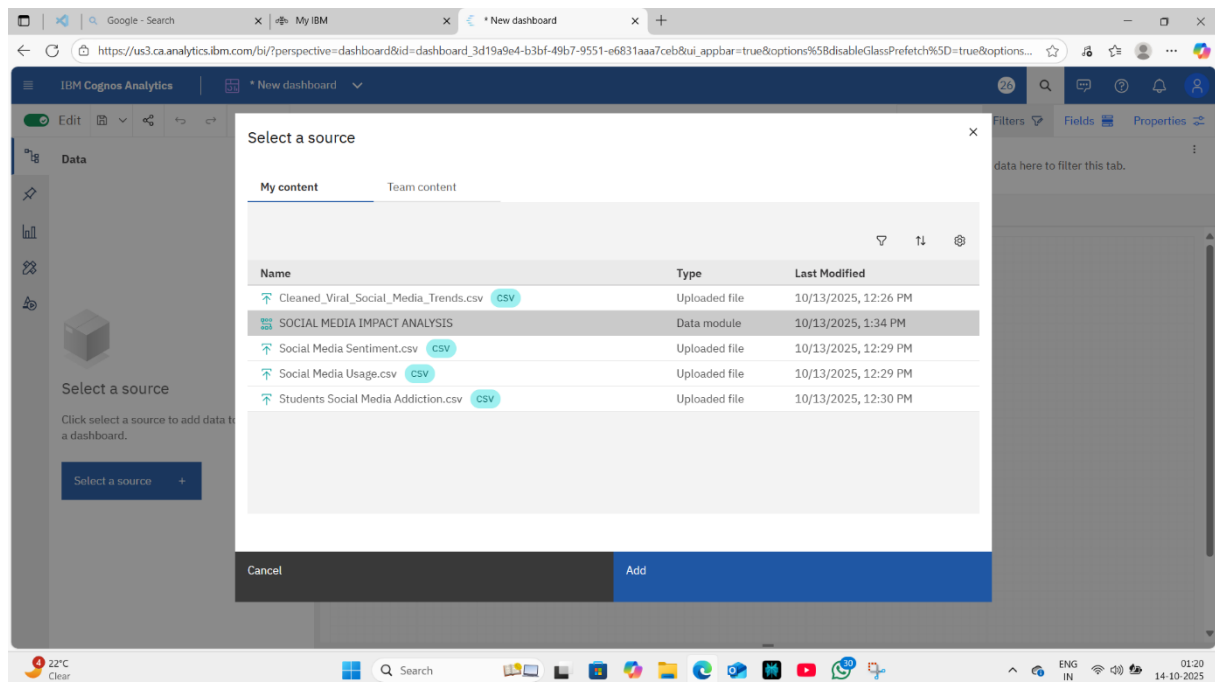
Step 2: Go to Click to Upload option and Select Files you want to Upload and Click on next



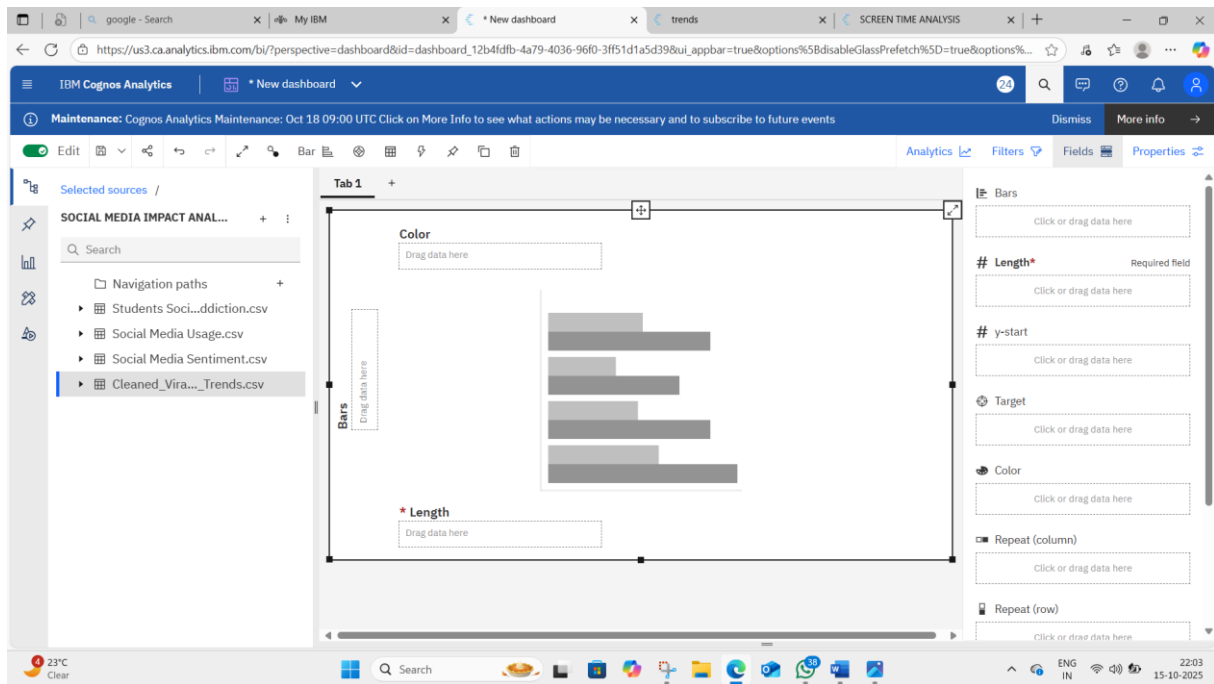
Step 3: Click on the DASHBOARD Option on next screen and Click on Create option.



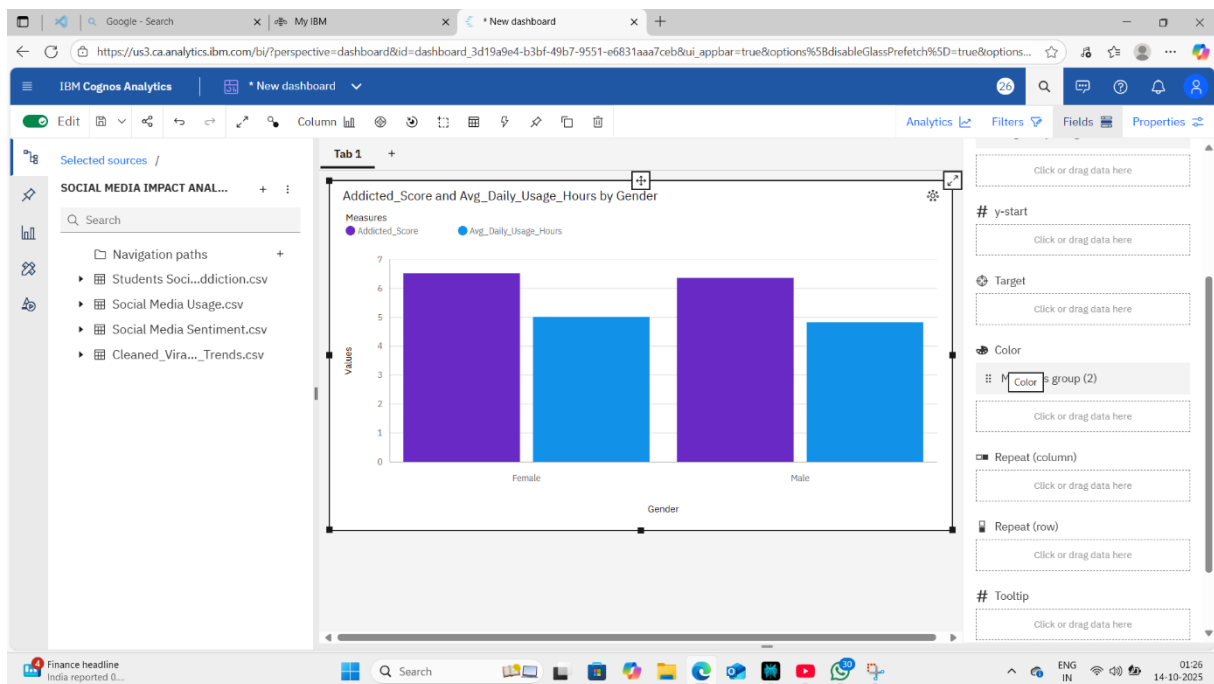
Step 4: Select Datasets to make a Dashboard



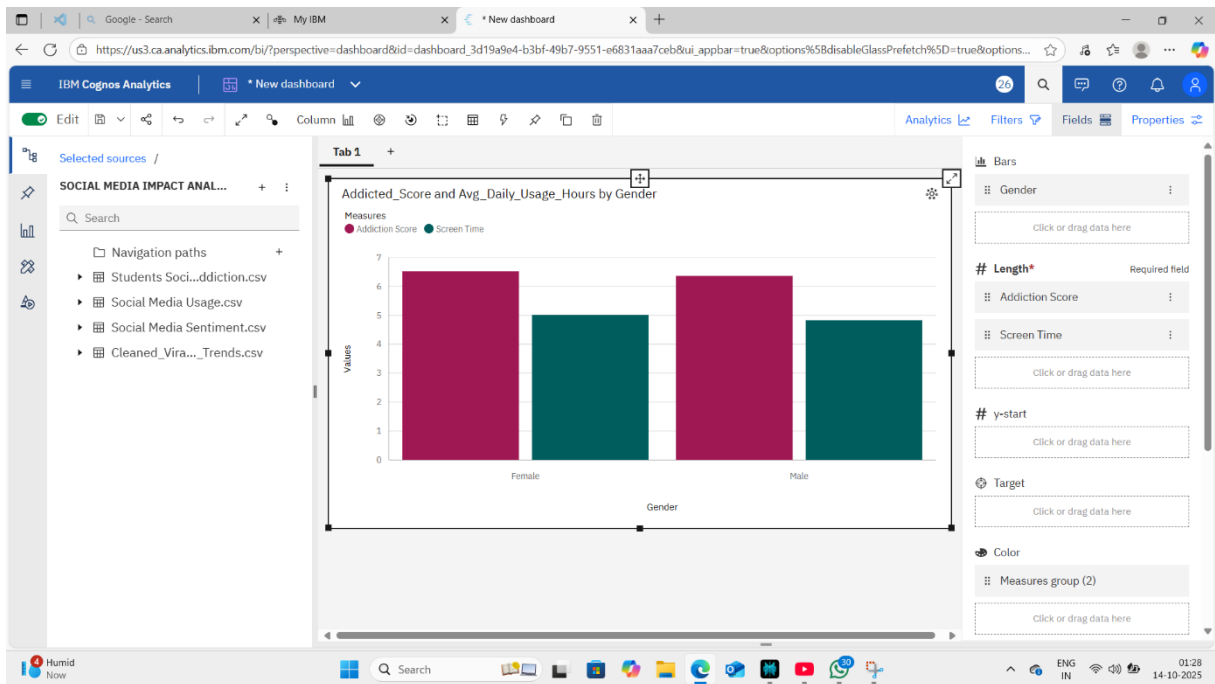
Step 5: Insert a Bar Chart Visualisation



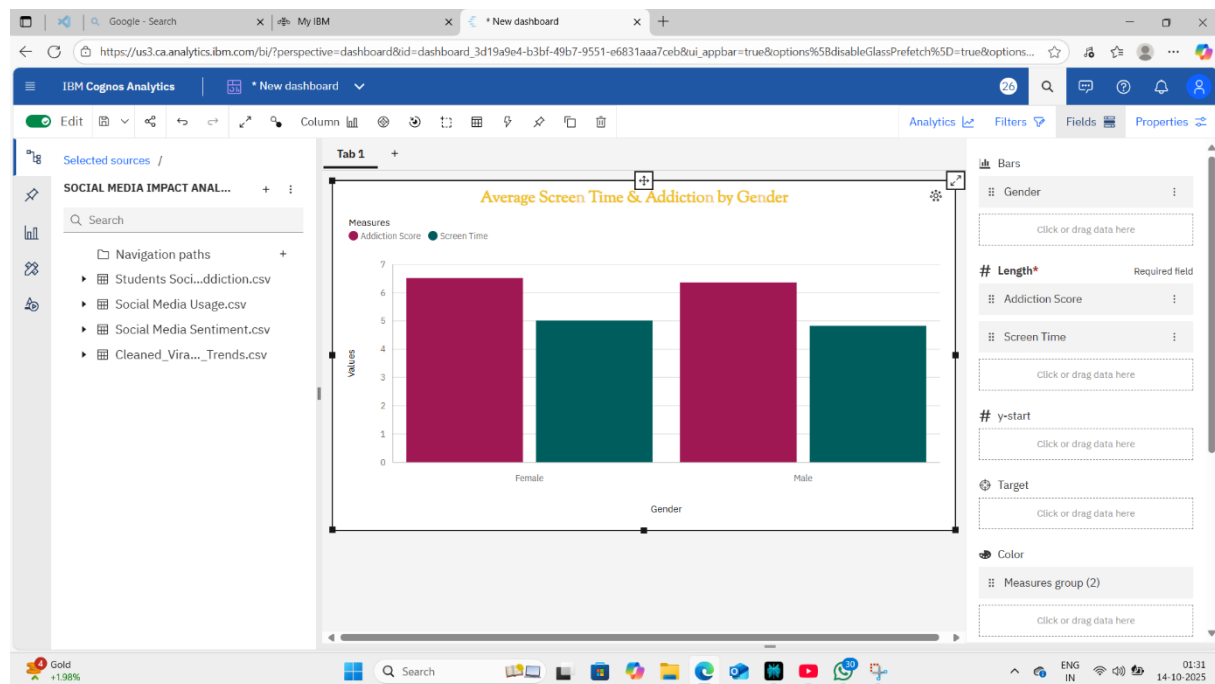
Step 6: ADD data to Bar Chart



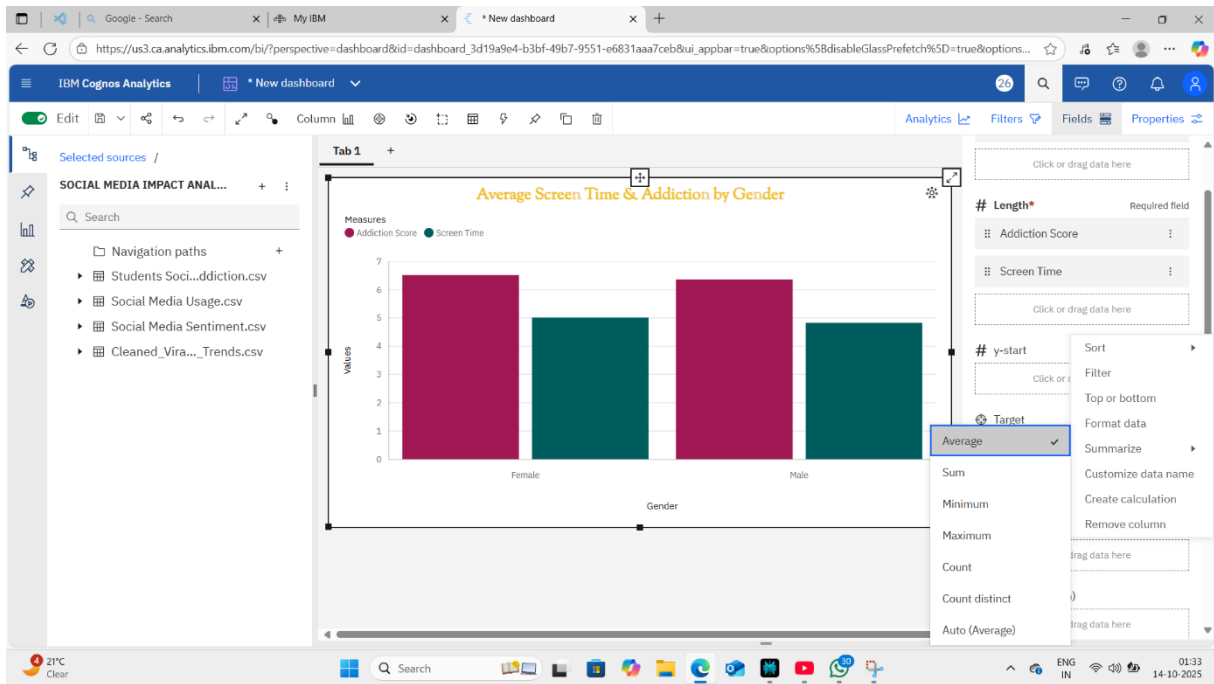
Step 7: Rename columns(for clarity) CHANGE avg_daily_usage to Screen Time



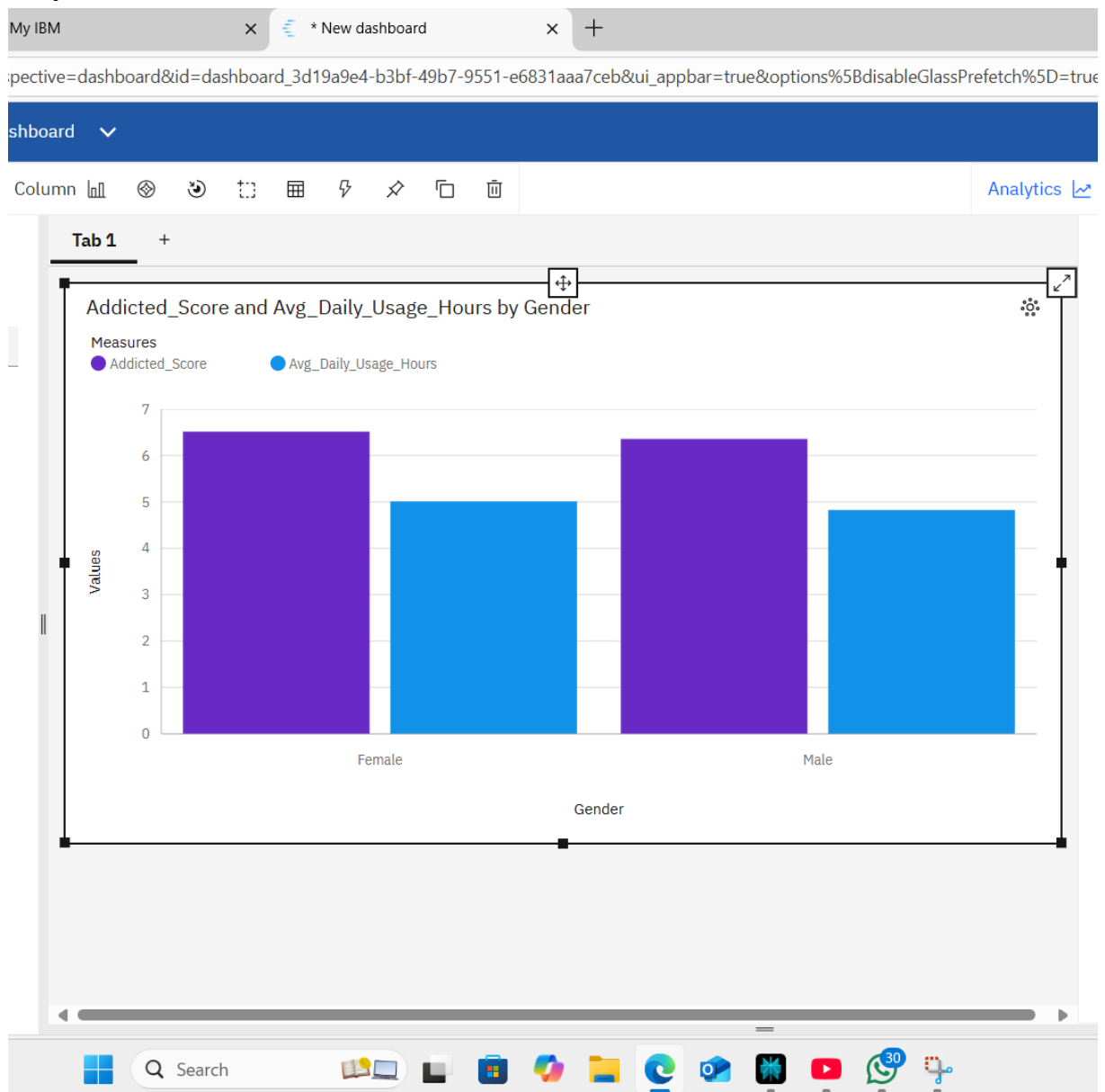
Step 8: ADD additional formatting to make it look good



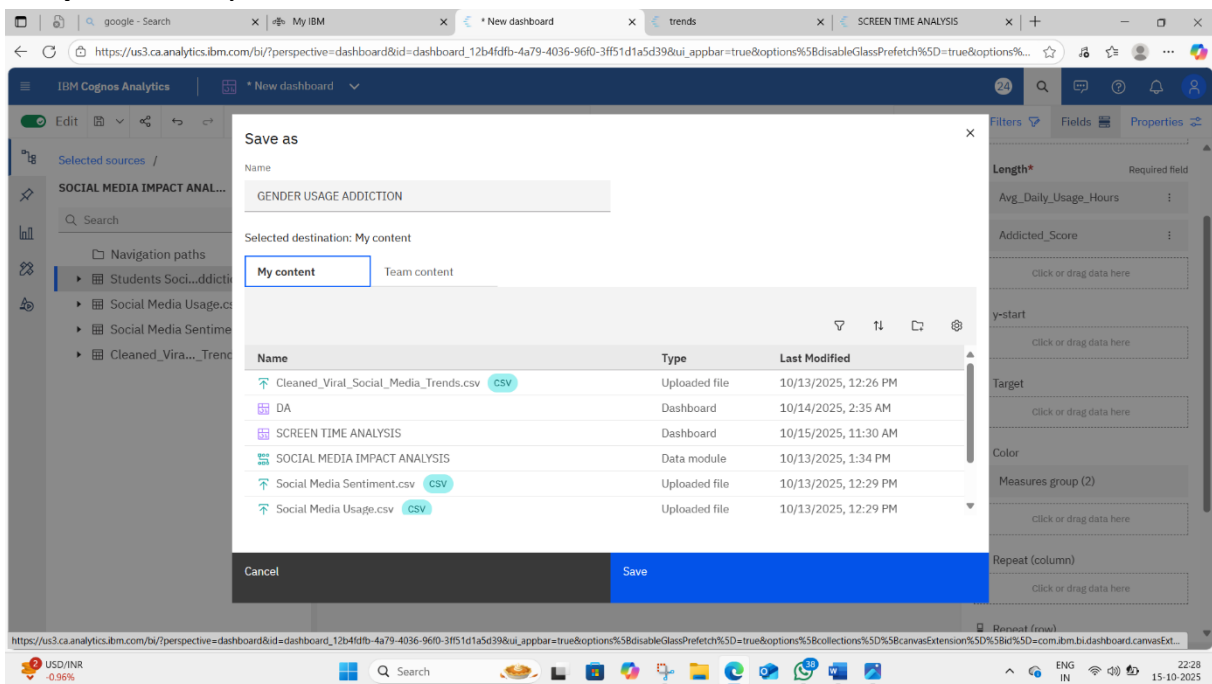
Step 9: Change the value type to “Average”.



Step 10: ADD COLOR customisation



Step 11: Save your work

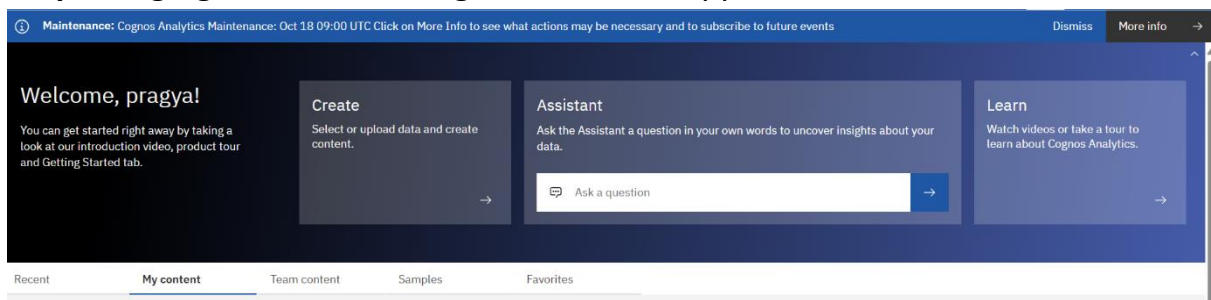


Project

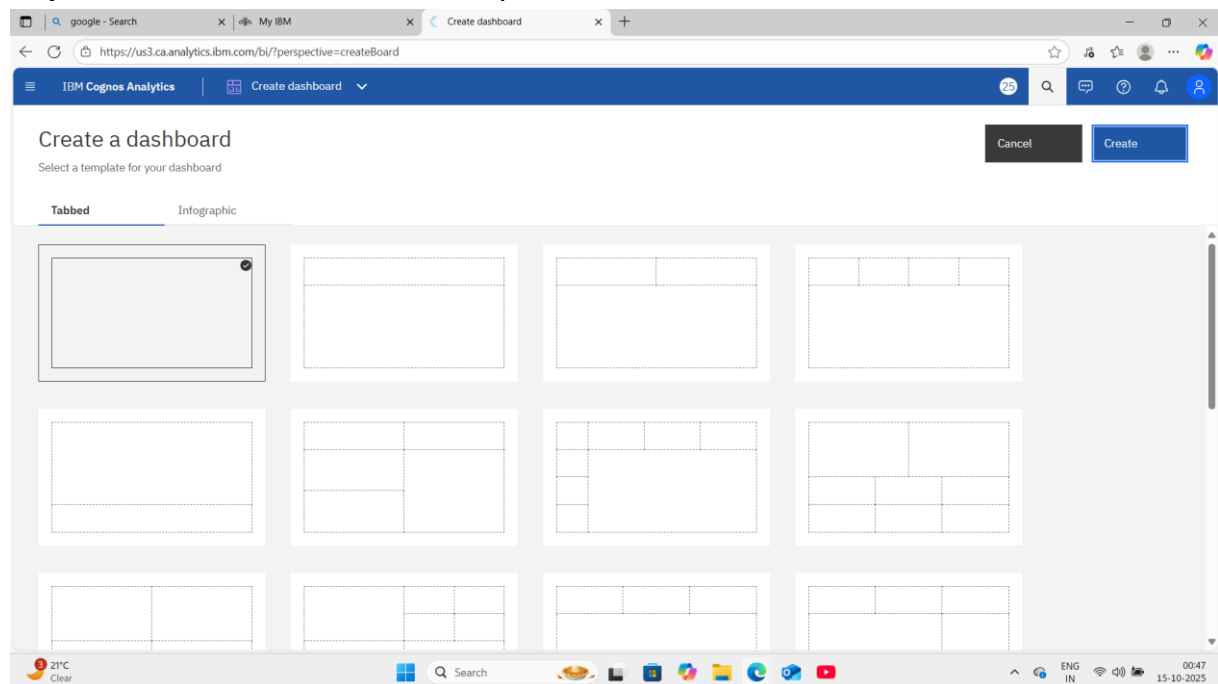
Problem Statement: Identify users with excessive screen time but low engagement rates

Solution: Make a table that shows those users with excessive screen time and low engagement rates.

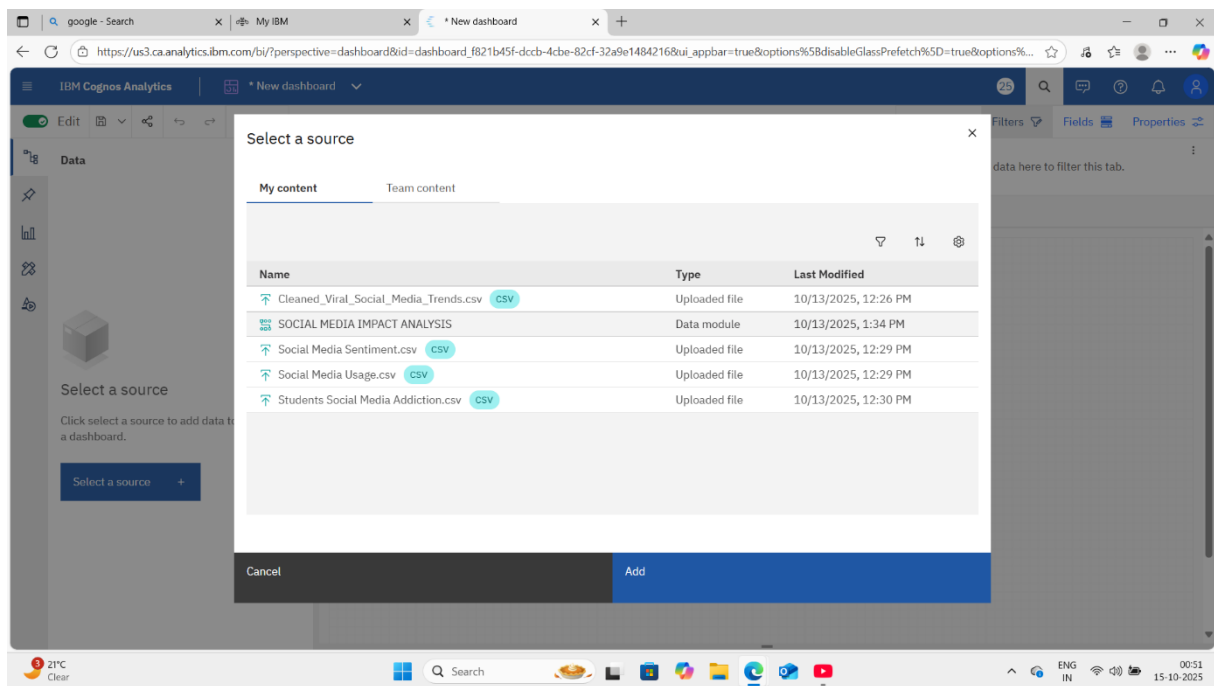
Step 1: logging to IBM following interface will appear



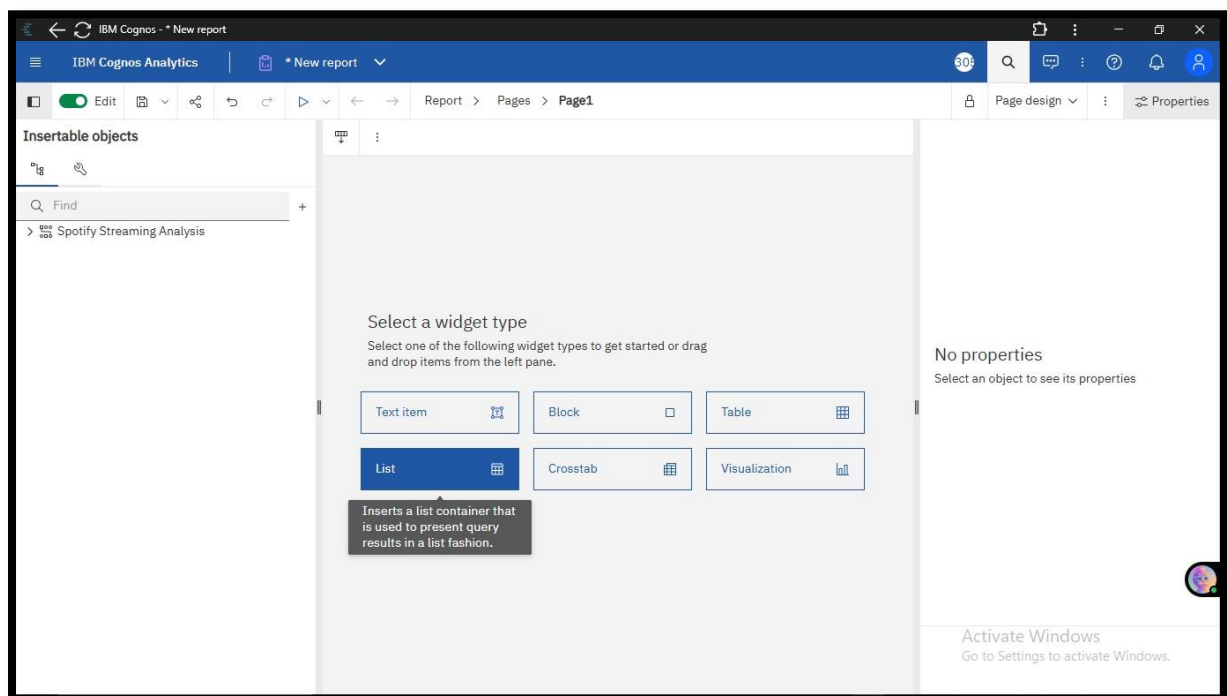
Step 2: Select Blank dashboard Option and Click on Create



Step 3: Select Data Module from My Content option and Click on Open Option.



Step 4: Select table Widget Type from the given Options



Step 5: Add data from data set one by one into columns fields

The image displays two screenshots of the IBM Cognos Analytics web interface, showing the process of creating a data table from CSV sources.

Top Screenshot:

- Left Panel (Selected sources):** Lists various CSV files under 'SOCIAL MEDIA IMPACT ANAL...'. The file 'Students So...diction.csv' is selected.
- Table 1:** A table titled 'Posts_Per_Day' with a single column 'Posts_Per_Day'. The data rows are: 1, 2, 4, 5, 6, 7, 8, 9, 10, 210.
- Right Panel (Columns):** Shows the 'Columns' section with 'Posts_Per_Day' listed as a required field.

Bottom Screenshot:

- Left Panel (Selected sources):** The same list of CSV files is shown. The file 'Social Medi...ntiment.csv' is selected.
- Table 1:** A table titled 'Daily_Usage_Time (minutes) and Posts_Per_Day' with two columns: 'Daily_Usage_Time (minutes)' and 'Posts_Per_Day'. The data rows are: (1, 9), (100, 2), (105, 4), (110, 6), (115, 3), (120, 4).
- Right Panel (Columns):** Shows the 'Columns' section with 'Daily_Usage_Time (minutes)' and 'Posts_Per_Day' listed as required fields.

The screenshot shows the IBM Cognos Analytics interface. On the left, a sidebar lists 'Selected sources' including 'SOCIAL MEDIA IMPACT ANAL...' and various CSV files. A search bar is present. The main area displays 'Tab 1' with a table titled 'User_ID, Daily_Usage_Time (minutes) and Posts_Per_Day'. The table has three columns: 'User_ID', 'Daily_Usage_Time (minutes)', and 'Posts_Per_Day'. The data is as follows:

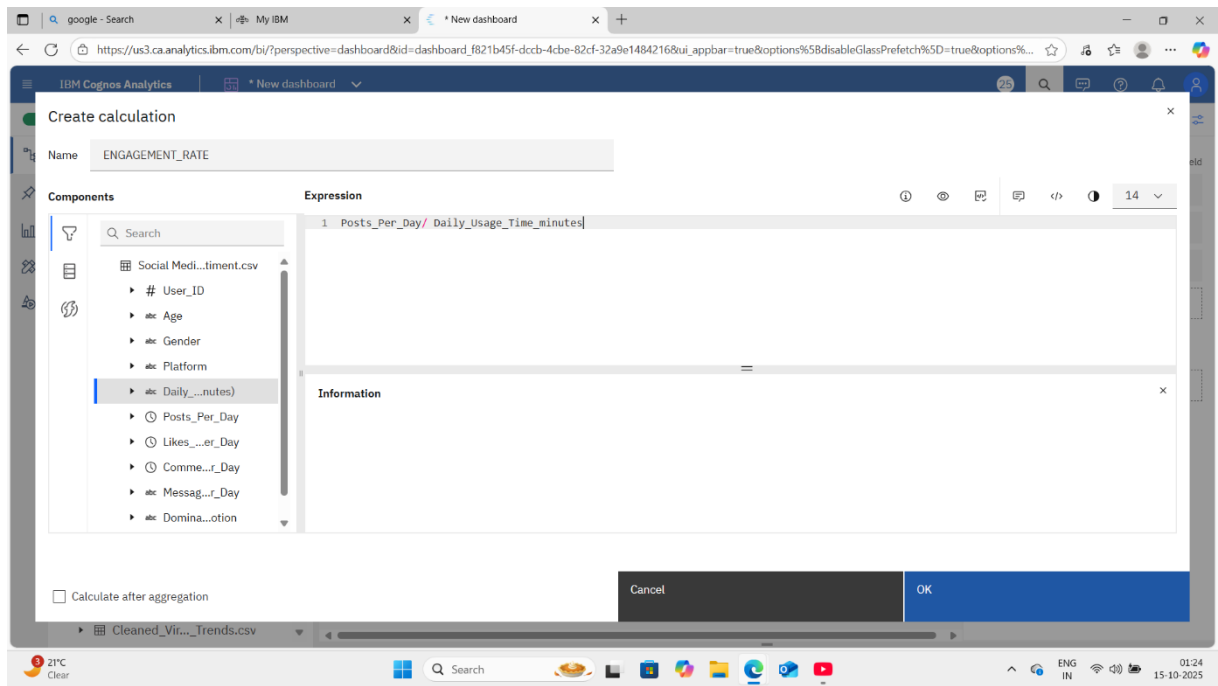
User_ID	Daily_Usage_Time (minutes)	Posts_Per_Day
5	45	1
10	170	5
13	75	6
20	130	4
23	60	2
27	1	9
28	160	6
54	145	4
64	120	4
69	105	3

On the right, the 'Columns' panel shows the selected fields: 'User_ID', 'Daily_Usage_Time (minutes)', and 'Posts_Per_Day'. Below it, the 'Local filters' panel is empty.

Step 6: ADD a calculation for measuring the engagement rate

This screenshot shows the same IBM Cognos Analytics interface as the previous one, but with a context menu open over the table. The menu options are: 'New', 'Calculation...', 'Filter...', 'Refresh members', and 'Properties...'. The 'Calculation...' option is highlighted with a blue border. The table data remains the same as in the previous screenshot.

Step 7: IN FORMULA BOX write:
[Engagement_count]/[screen_time]



Step 8: Make a new column(engagement rate)

The screenshot shows the IBM Cognos Analytics interface. On the left, the 'Selected sources' pane lists various data sources, including 'SOCIAL MEDIA IMPACT ANAL...' and 'ENGAGEMENT_RATE'. The main workspace displays a table with columns: 'User_ID', 'Daily_Usage_Time (minutes)', 'Posts_Per_Day', and 'ENGAGEMENT_RATE'. The table contains 10 rows of data. On the right, the 'Columns' pane shows the 'ENGAGEMENT_RATE' column being added to the table.

User_ID	Daily_Usage_Time (minutes)	Posts_Per_Day	ENGAGEMENT_RATE
5	45	1	0.02222222222222223
10	170	5	0.029411764705882353
13	75	6	0.08
20	130	4	0.03076923076923077
23	60	2	0.03333333333333333
27	1	9	9.0
28	160	6	0.0375
30	90	2	0.02222222222222223
54	145	4	0.027586206896551724
64	120	4	0.03333333333333333
69	105	3	0.02857142857142857

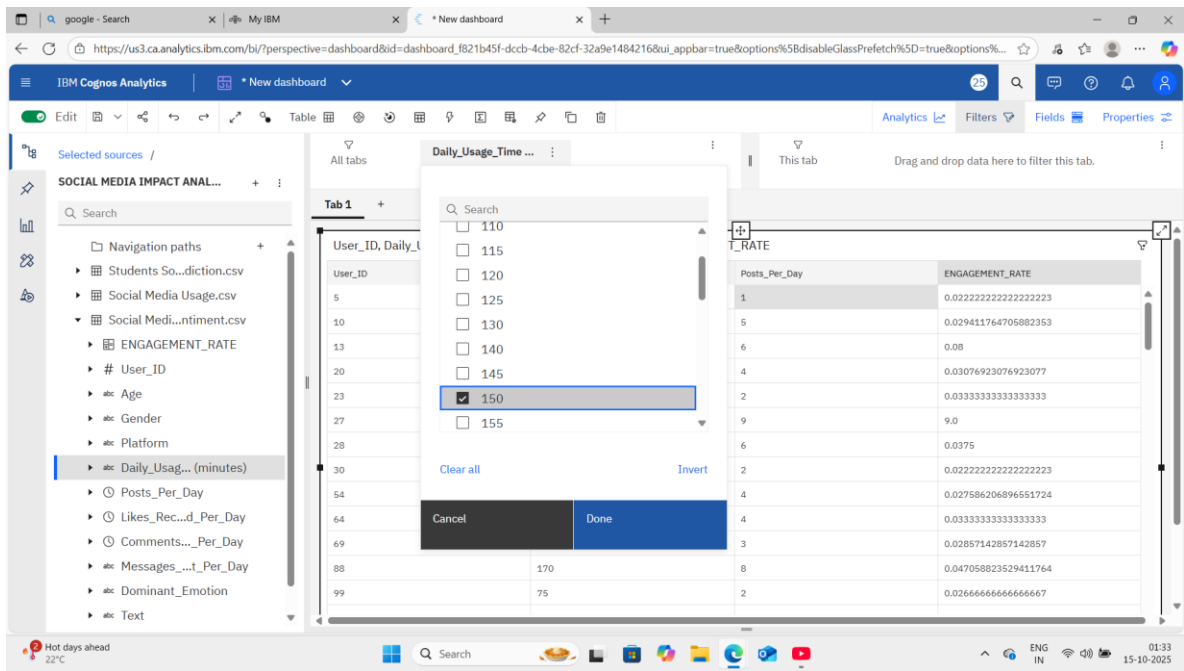
Step 9: Add Filters to find “problem users”

The screenshot shows the IBM Cognos Analytics interface with filters applied to the table. The 'Daily_Usage_Time (minutes)' column is selected in the 'Selected sources' pane. The main workspace displays the table with the same columns as before. The 'Columns' pane on the right shows the 'Daily_Usage_Time (minutes)' column being added to the table. The table contains 10 rows of data.

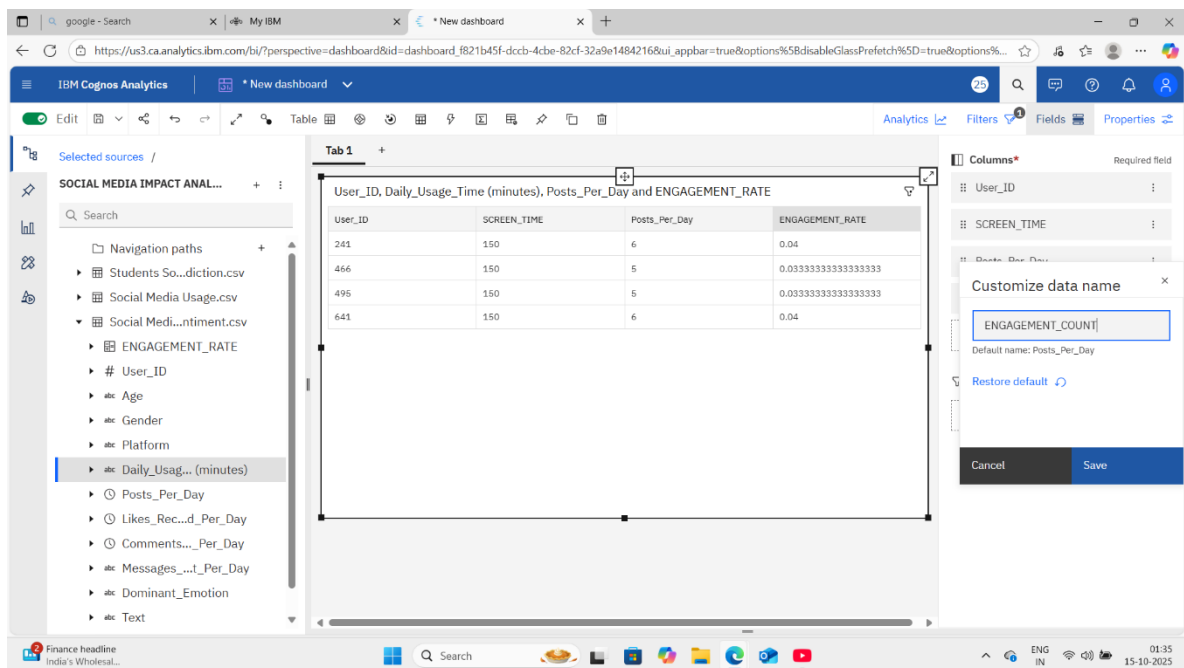
User_ID	Daily_Usage_Time (minutes)	Posts_Per_Day	ENGAGEMENT_RATE
5	45	1	0.02222222222222223
10	170	5	0.029411764705882353
13	75	6	0.08
20	130	4	0.03076923076923077
23	60	2	0.03333333333333333
27	1	9	9.0
28	160	6	0.0375
30	90	2	0.02222222222222223
54	145	4	0.027586206896551724
64	120	4	0.03333333333333333
69	105	3	0.02857142857142857
88	170	8	0.047058823529411764
99	75	2	0.02666666666666667

Page

Step 11: ADD filter condition to find the users using more than 150 minutes.



Step 12: Customise data name for better understanding



Step 13: Highlight them with color in properties icon

The screenshot shows the IBM Cognos Analytics interface. A table titled "SCREEN TIME VS ENGAGEMENT RATE" is displayed with the following data:

User_ID	SCREEN_TIME	ENGAGEMENT_COUNT	ENGAGEMENT_RATE
241	150	6	0.04
466	150	5	0.0333333333333333
495	150	5	0.0333333333333333
641	150	6	0.04

The table is highlighted in yellow. The right sidebar shows the "Visualization properties" panel with the "Color" property set to "#F1C21B".

Step 14: Save the dashboard

The screenshot shows the IBM Cognos Analytics interface with a "Save as" dialog box open. The dialog box has a "Name" field with the text "SCREEN TIME ANALYSIS" and a "Selected destination" of "My content". Below the dialog box, a list of existing content items is shown:

Name	Type	Last Modified
Cleaned_Viral_Social_Media_Trends.csv	CSV	10/13/2025, 12:26 PM
DA	Dashboard	10/14/2025, 2:35 AM
SCREEN TIME ANALYSIS	Dashboard	10/15/2025, 11:30 AM
SOCIAL MEDIA IMPACT ANALYSIS	Data module	10/13/2025, 1:34 PM
Social Media Sentiment.csv	CSV	10/13/2025, 12:29 PM
Social Media Usage.csv	CSV	10/13/2025, 12:29 PM

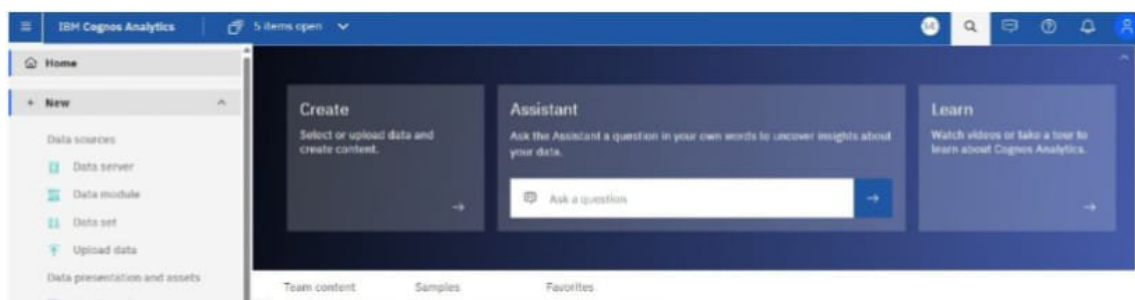
The dialog box has "Cancel" and "Save" buttons at the bottom.

Project

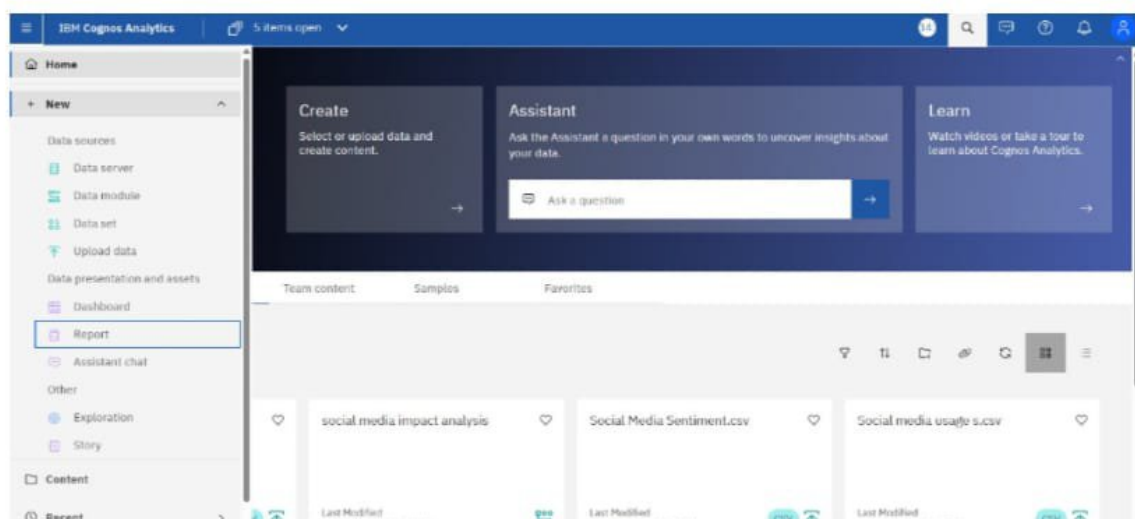
Identify the top 10 most engaging social media posts or trends

Solution: create a list using data module

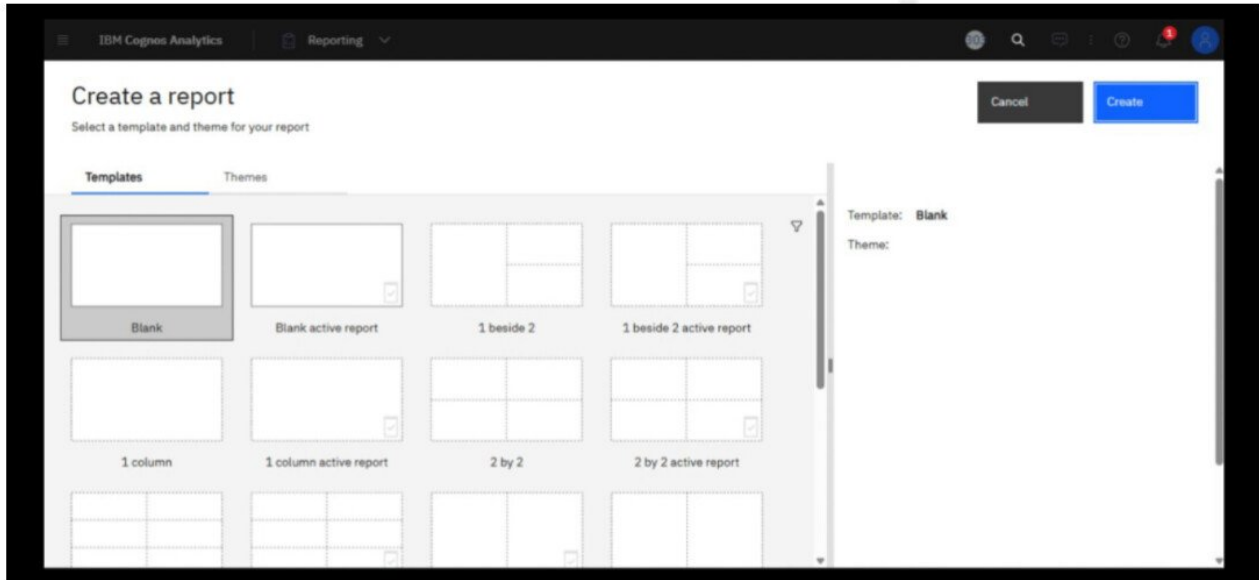
Step1 - Logging to IBM and followed interface will be opened



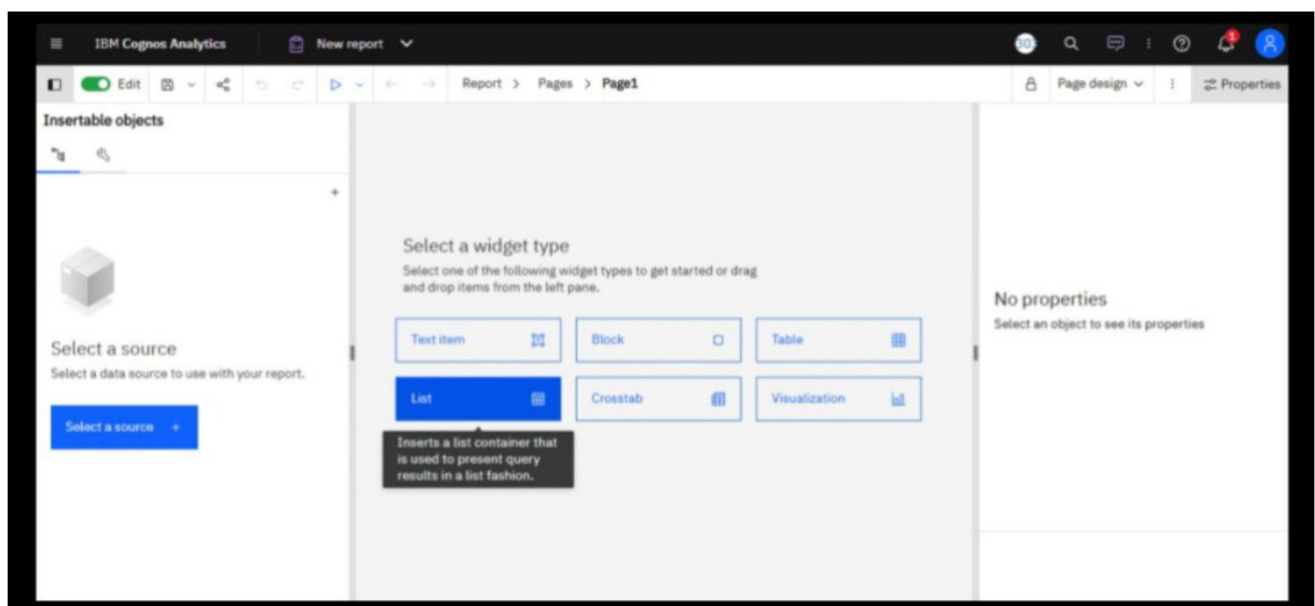
Step2 - Click three line icon and select new ->Report option.



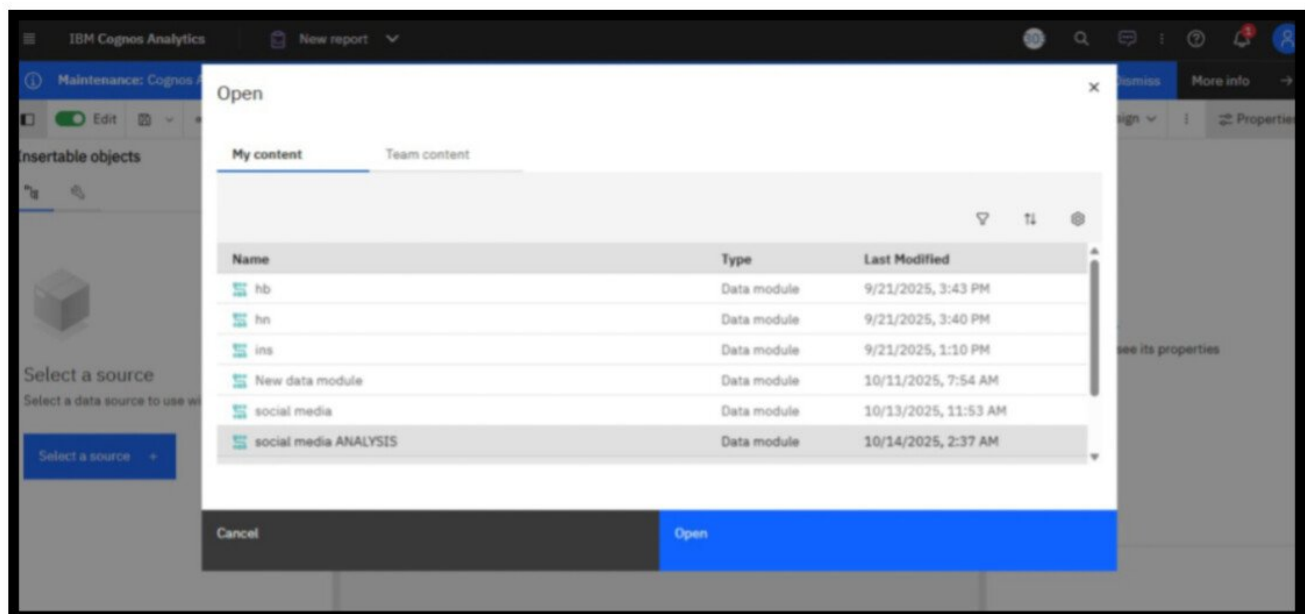
Step3- Click on Blank Report Option and then click on create option.



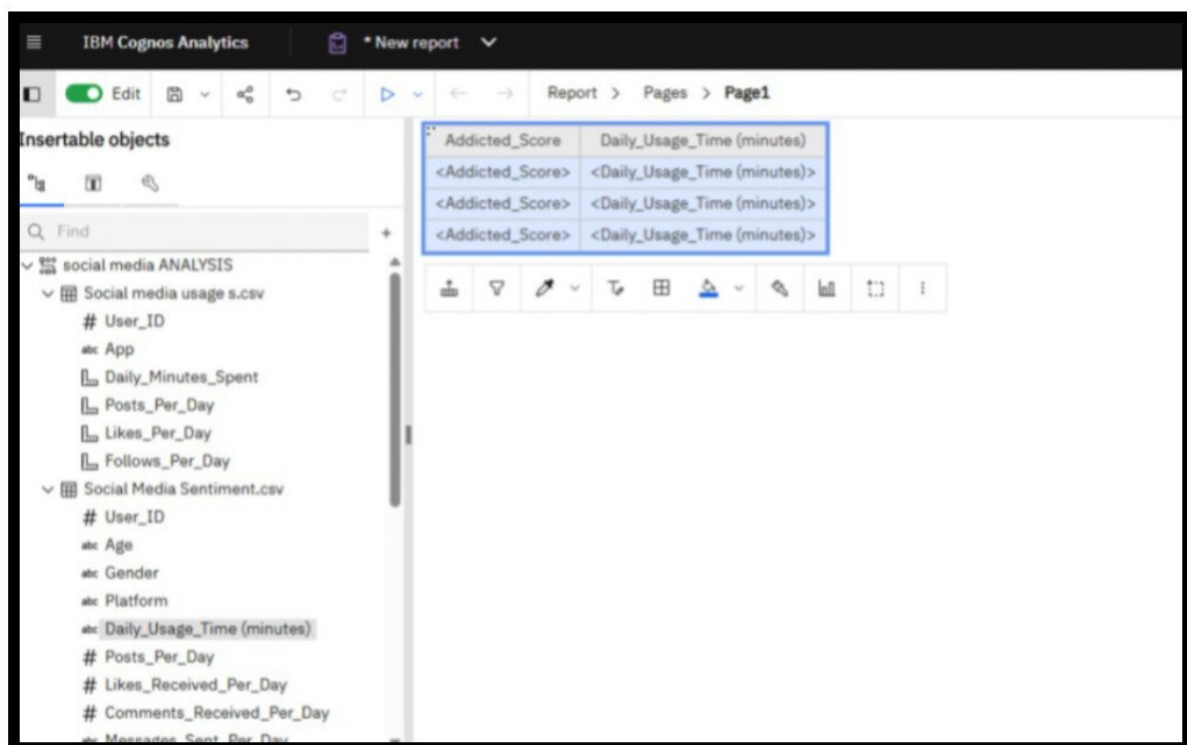
Step4- Select a List Report Option.



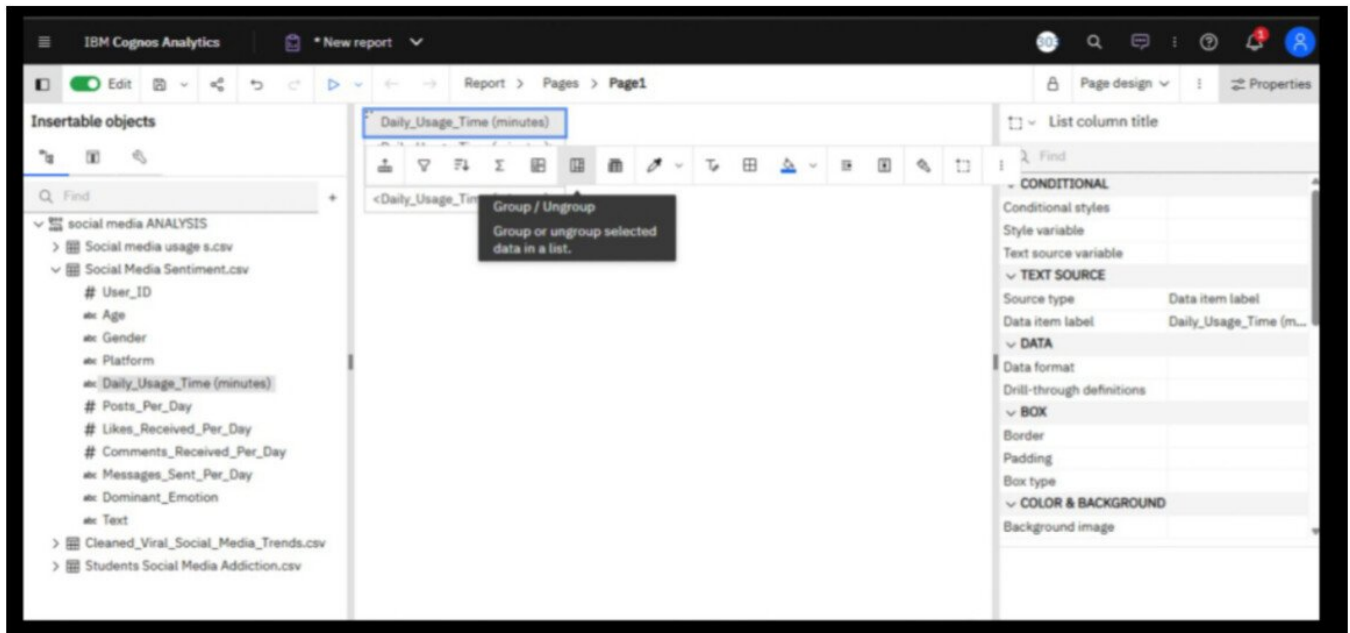
Step 5- Select Data Module from My Content option and Click an Open .



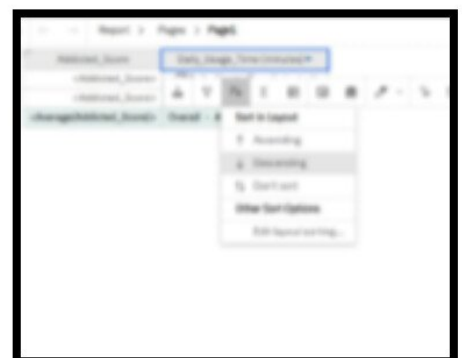
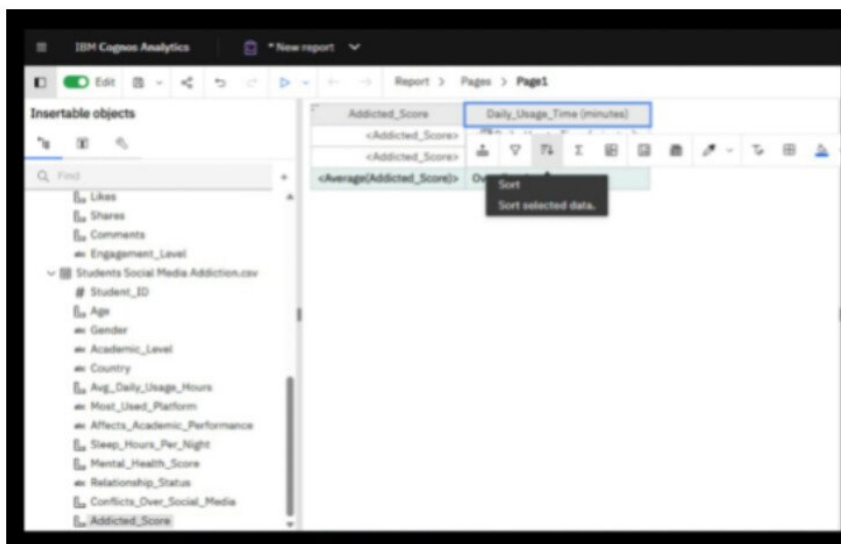
Step 6- Drag Daily Usage Time (minutes) and Addicted Score into the List Report.



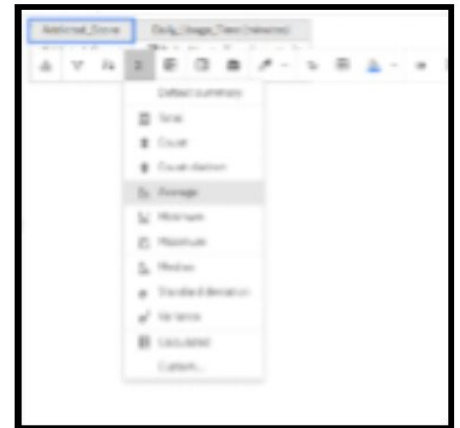
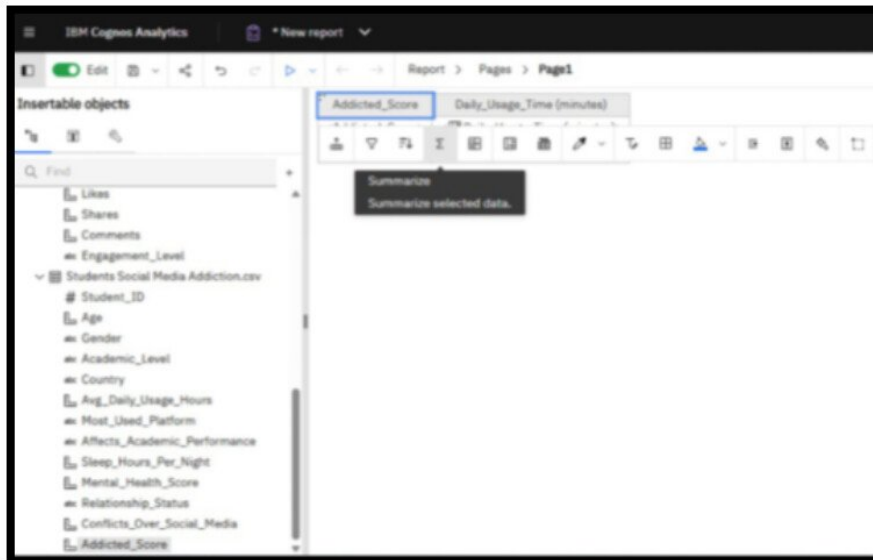
Step7- Click on Group Option .



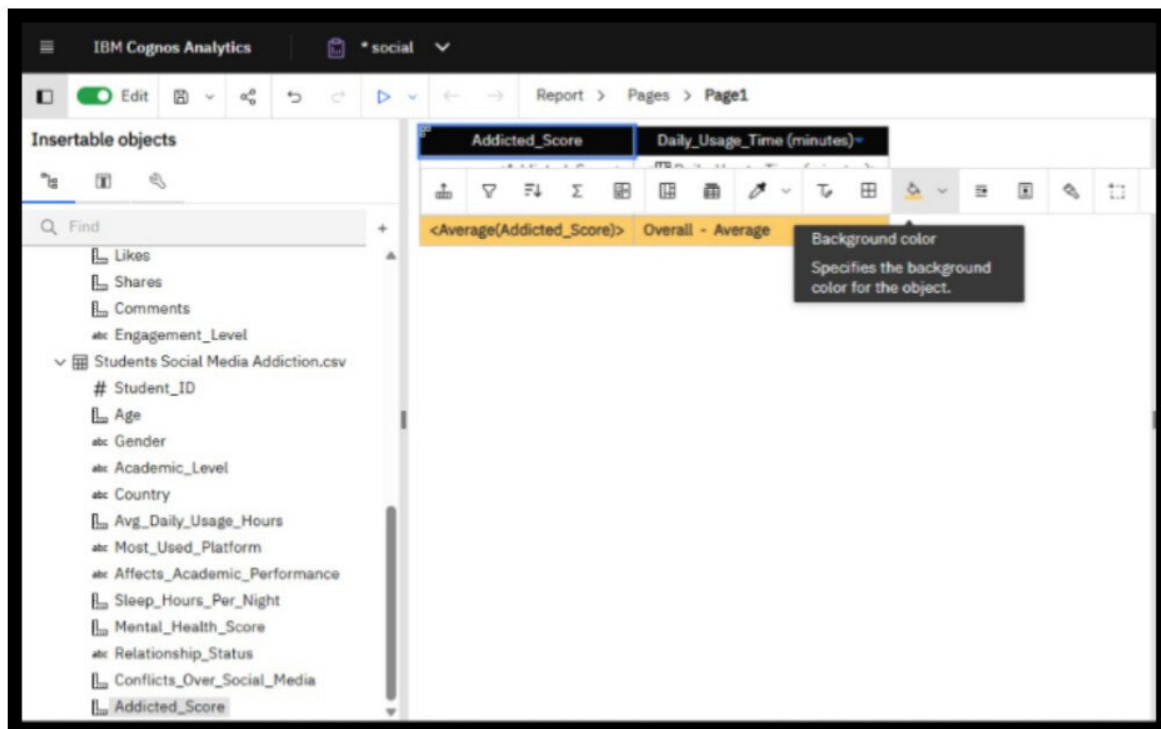
Step 8- Sort or Arrange the Data in Descending Order.



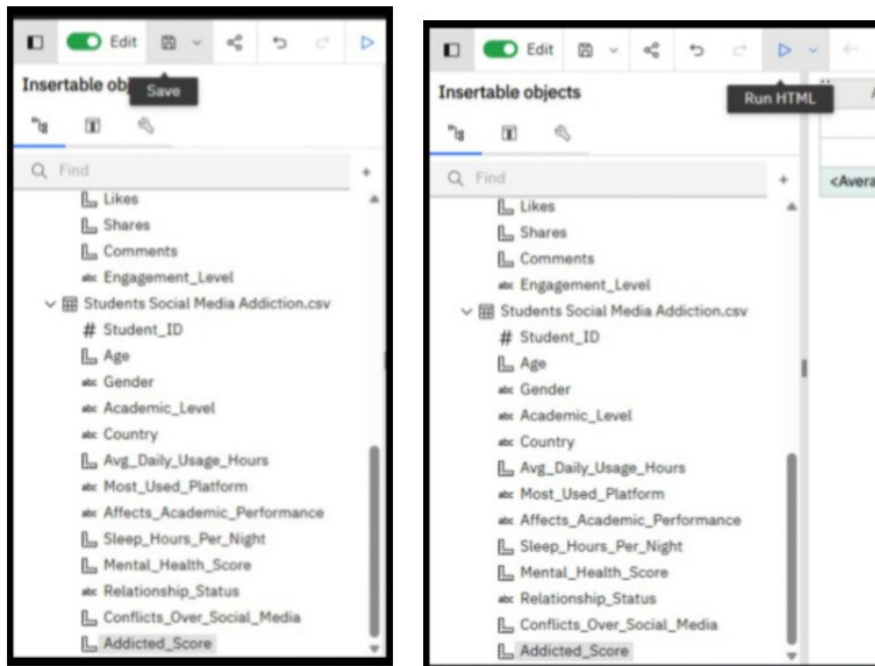
**Step 9- Drag Addiction Score into the list .
Right Click the Addiction Score column-> Summarize-> Average .**



Step 10- Click on Background color for changing the color .



Step 11- Click Save -> Run



You'll see table like this ->

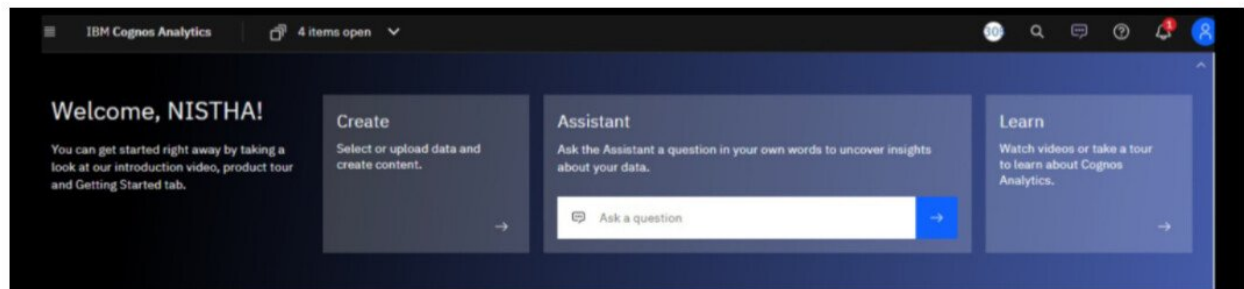
Addicted_Score	Daily_Usage_Time (minutes)
5.62202381	95
5.97415186	90
5.10434783	85
5.66666667	80
5.80184006	75
5.5	72
5.58436214	70
5.66666667	65
5.82252142	60
5.1959799	55
6.55421687	52
5.28802589	50
5.53773585	45
5.78481013	40
3.80952381	30
6.55421687	200
6.56590684	190
6.55421687	180
6.55421687	170
6.55421687	165

Project

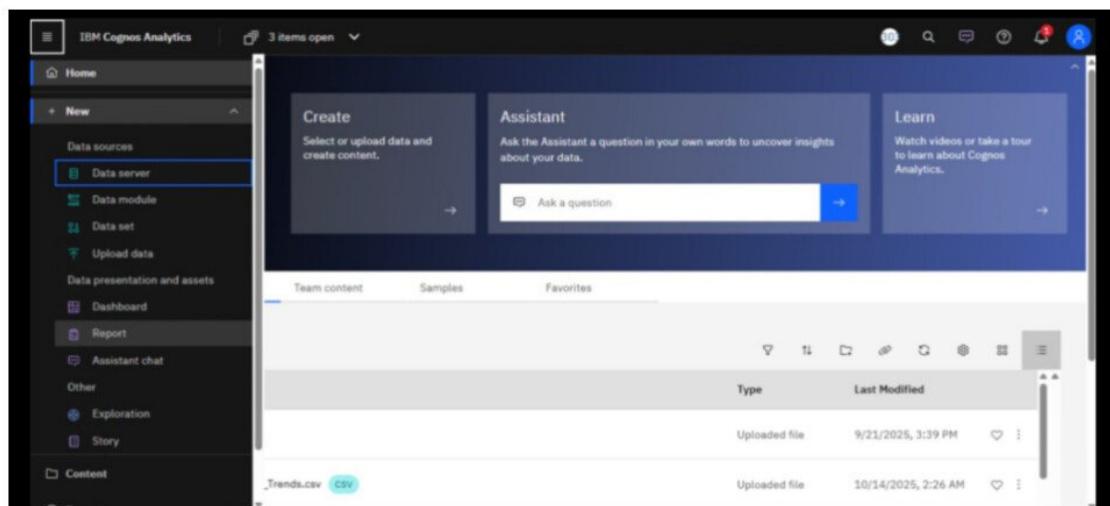
Problem statement: Examine the correlation between daily usage time and addiction score.

Solution: Created a List using Data module

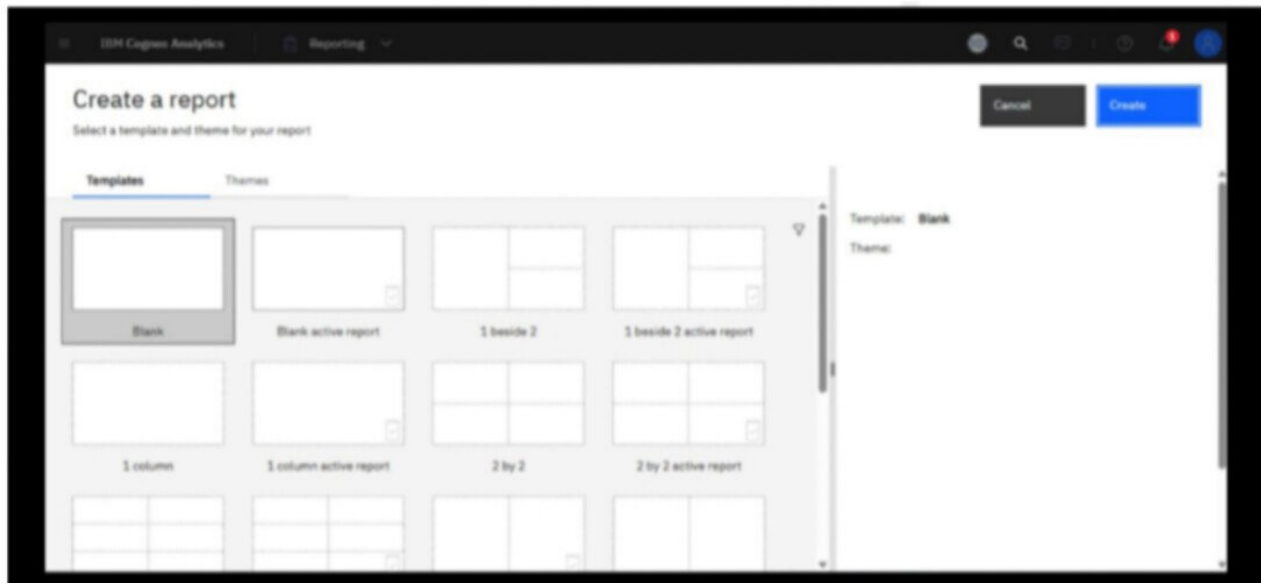
Step1- Login to IBM and followed interface will be opened .



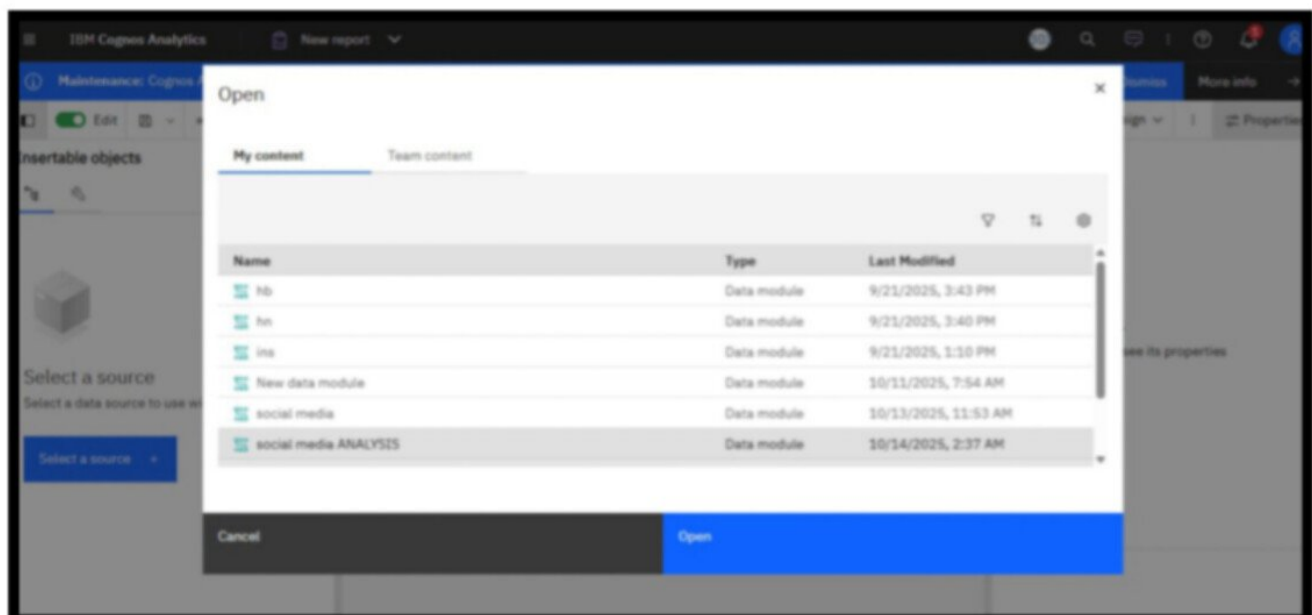
Step2- Click three-line icon and select New -> Report option.



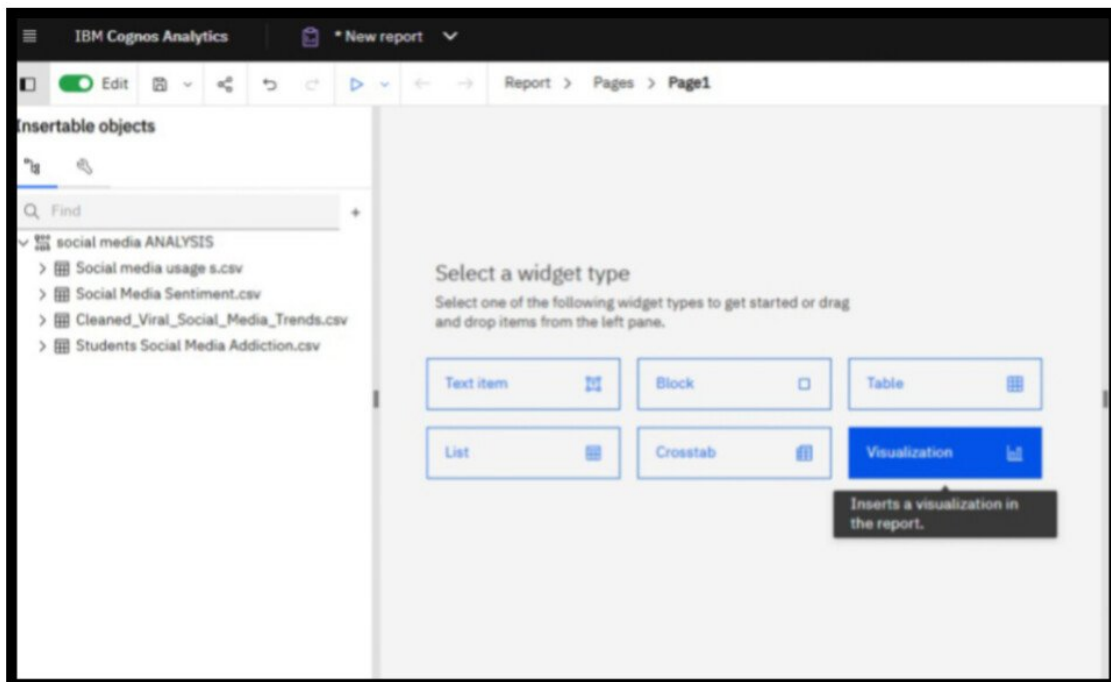
Step3- Click on Blank Report Option and then click on create option.



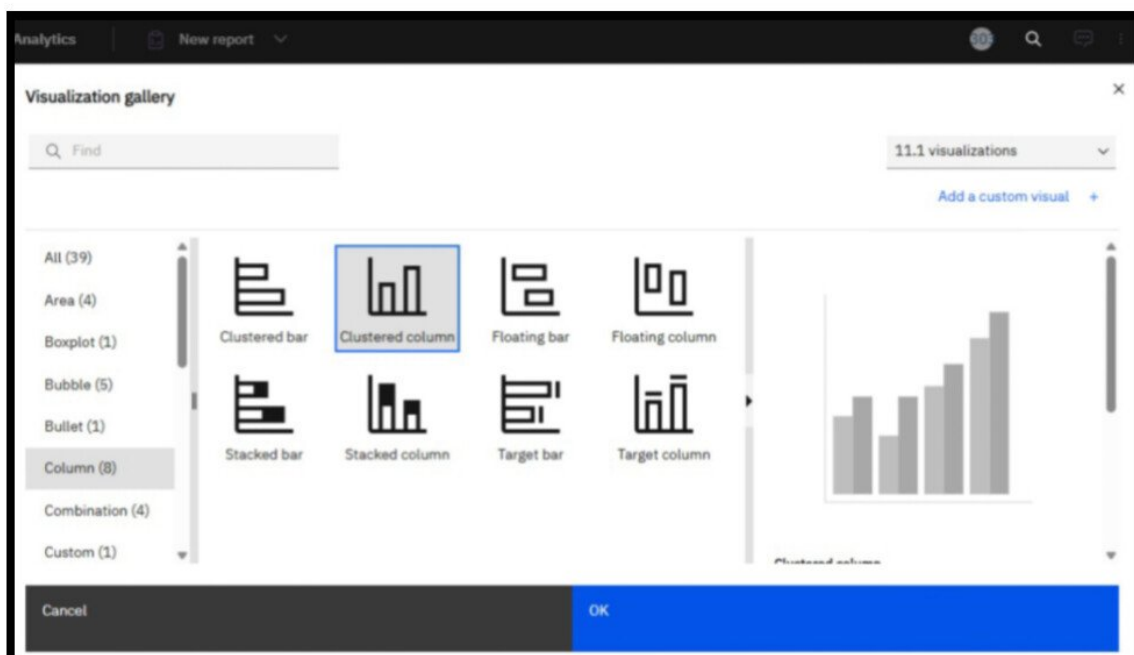
Step 4- Select Data Module from My Content and click an open



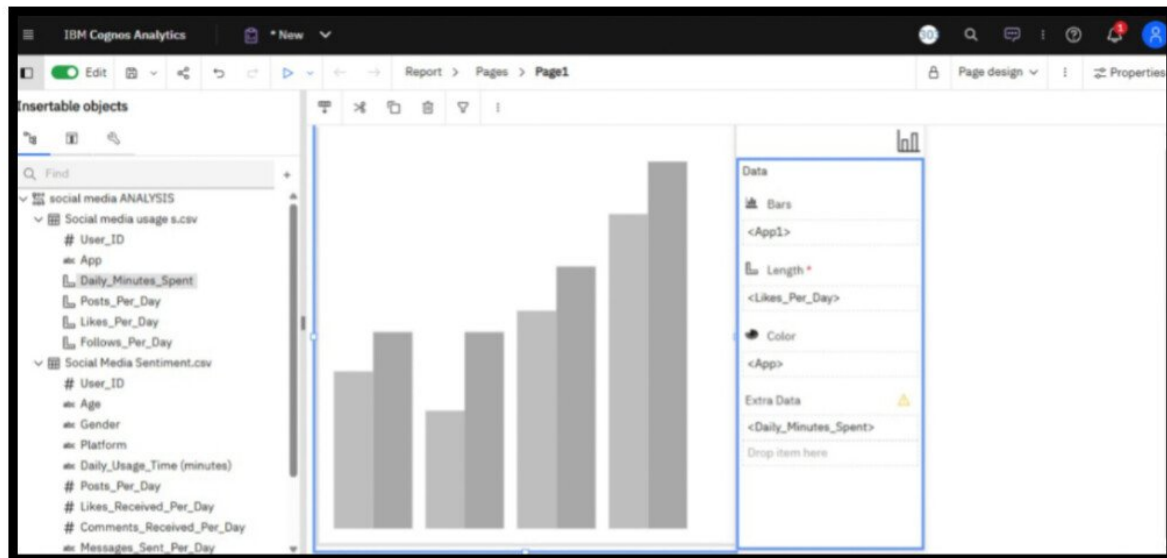
Step 5- Select Visualization widget from the given option.



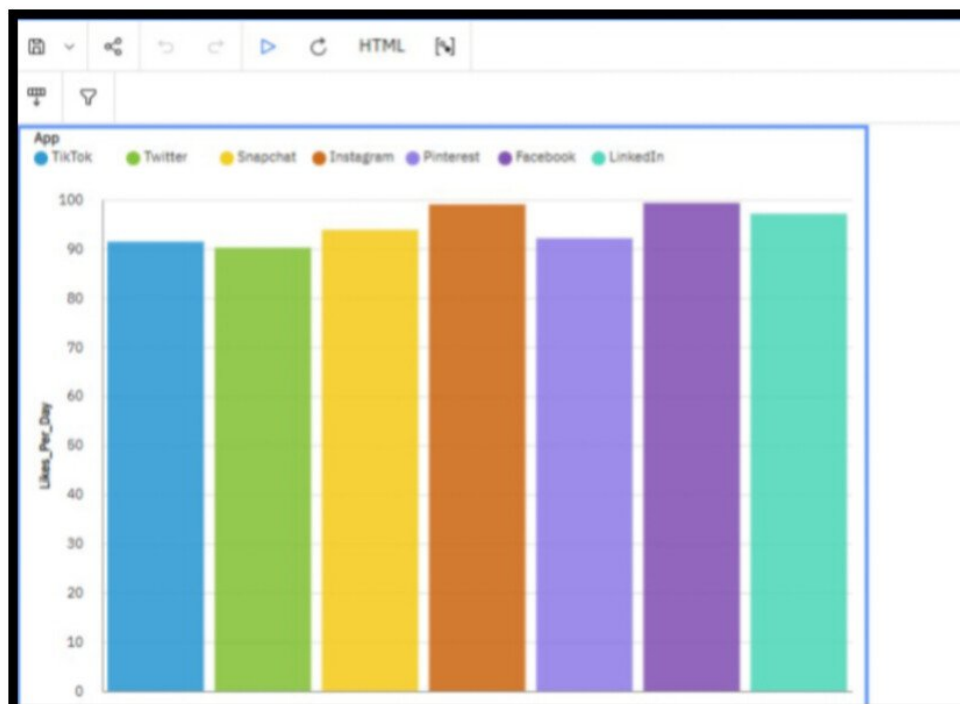
Step 6- Select a Clustered Columns .



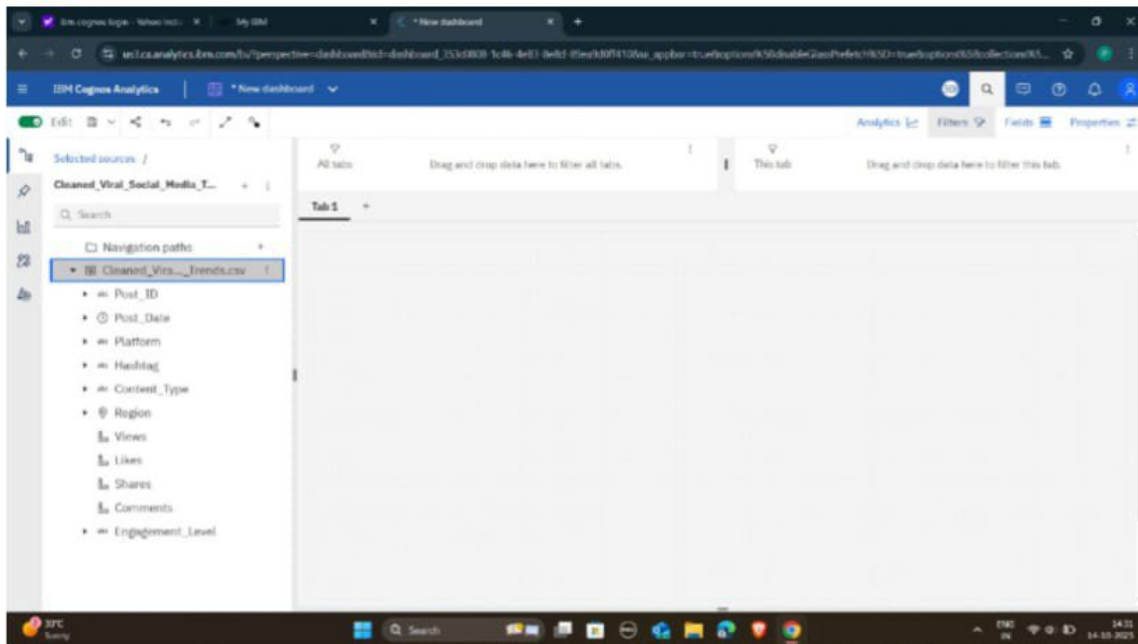
Step 7- Drag 'App' from Social Media Usage into Bars and Color, and <Likes_Per_Day> in length.



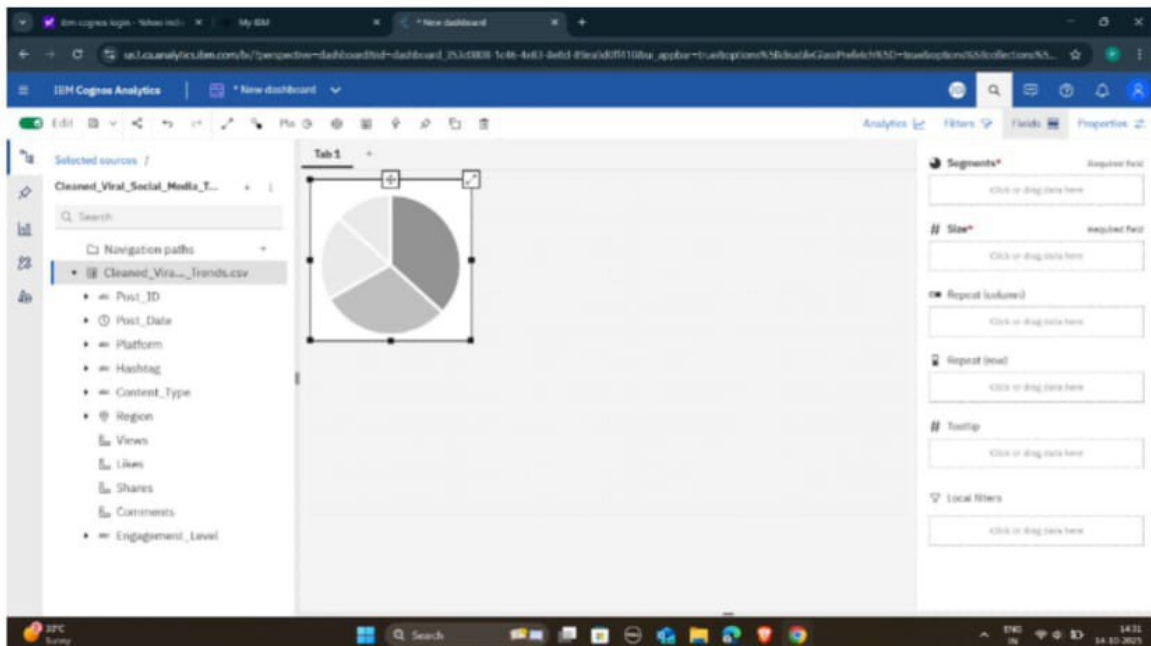
Step 8- Save and Run the report and following output will be generated.



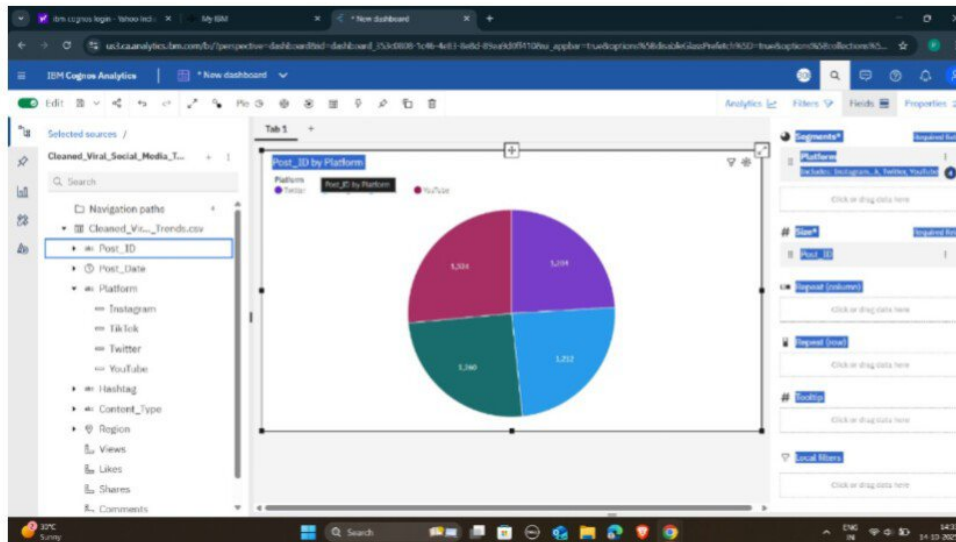
Step3: - Insert the Pie chart visualization onto the dashboard



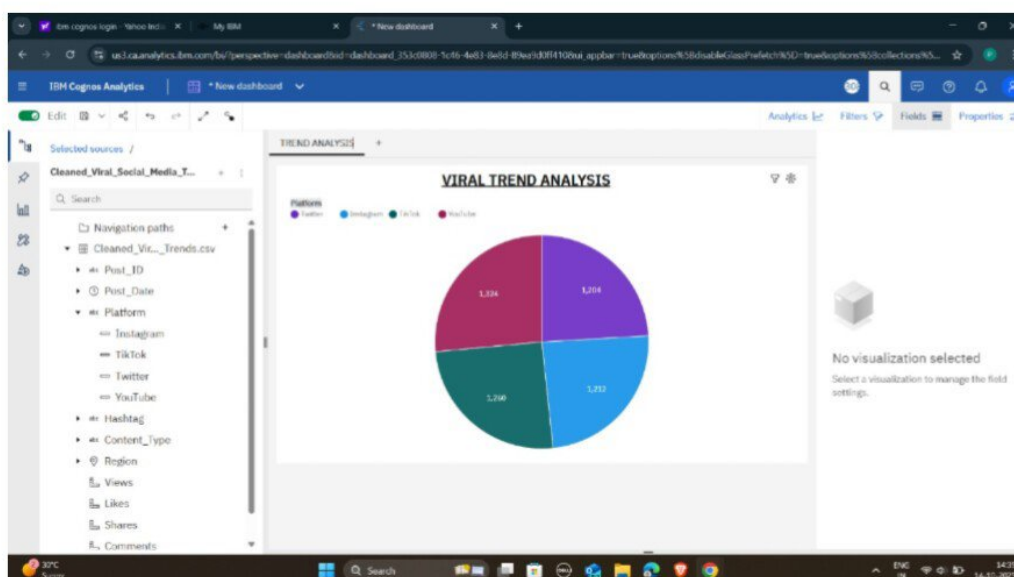
Step4: - Configure the chart by dragging Platform to Segments and Post_ID to Size (Count)



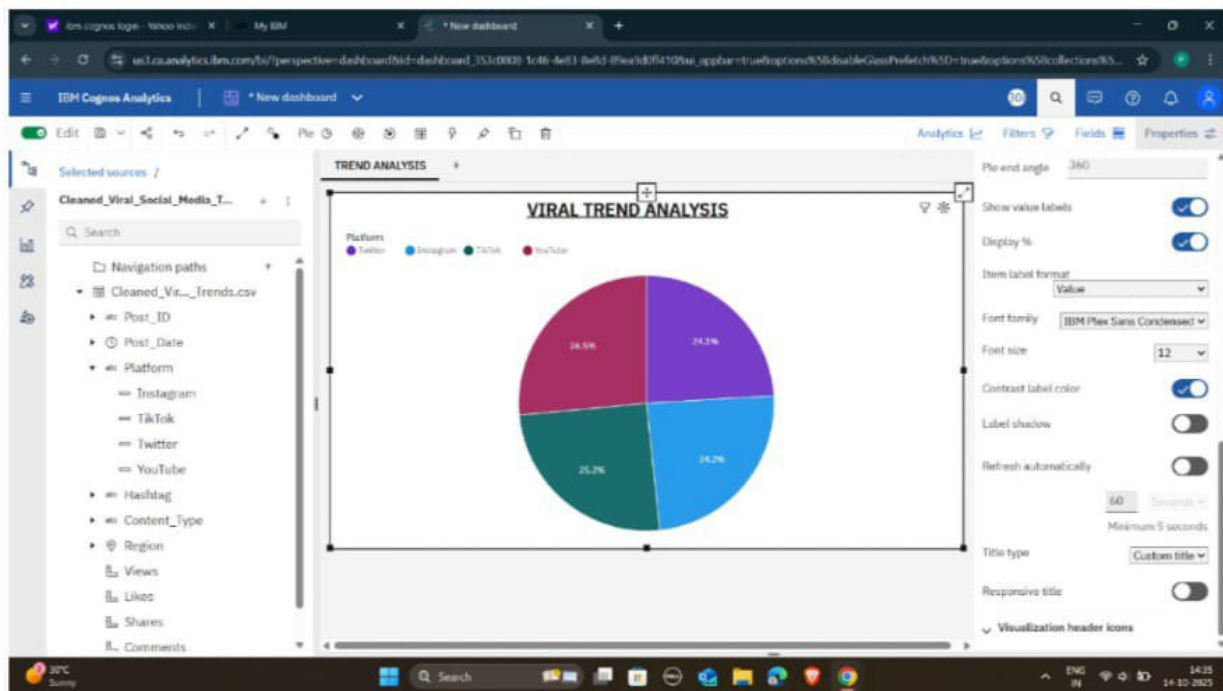
Step5: - Set the tab name to TREND ANALYSIS and the chart title to VIRAL TREND ANALYSIS



Step6: - Enable Display % in the properties pane I to show percentage values on the chart



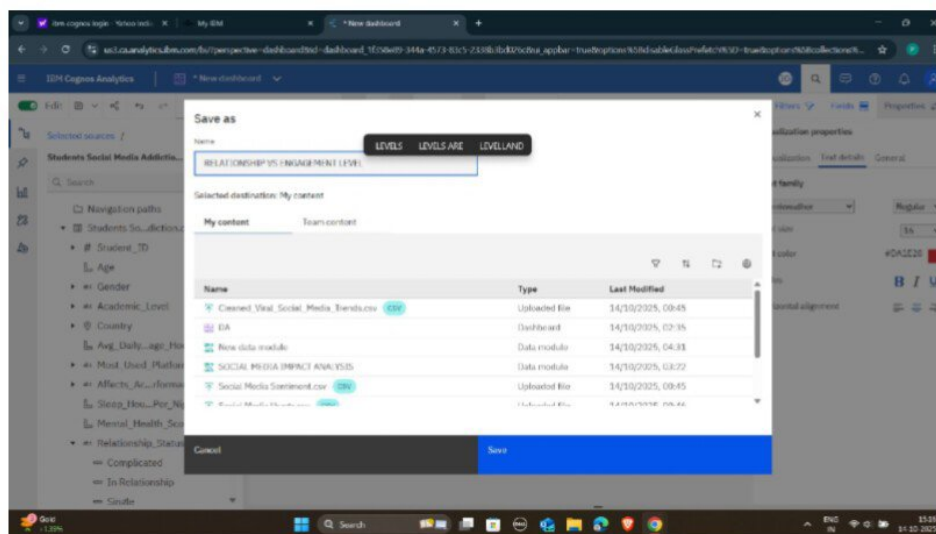
Step7: - Save the completed dashboard with the name TREND ANALYSIS to My content



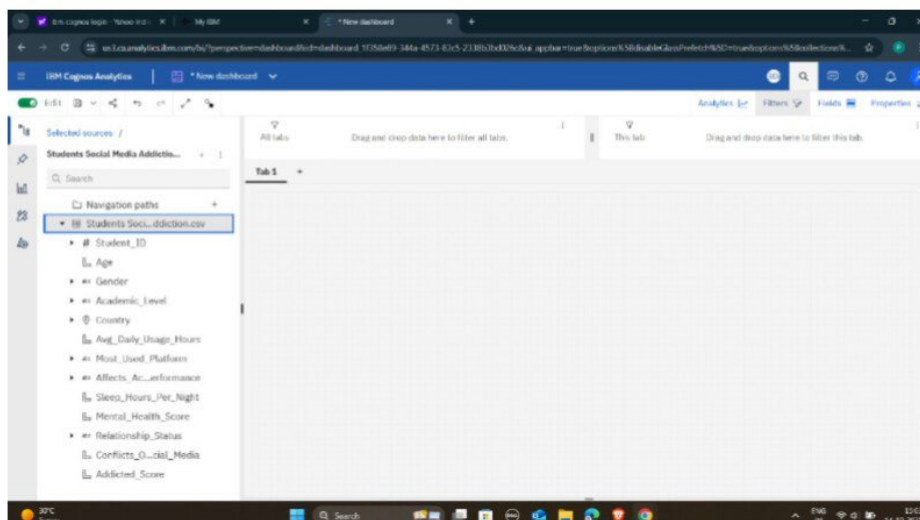
Project

Study how relationship equality changes with increasing social media engagement.

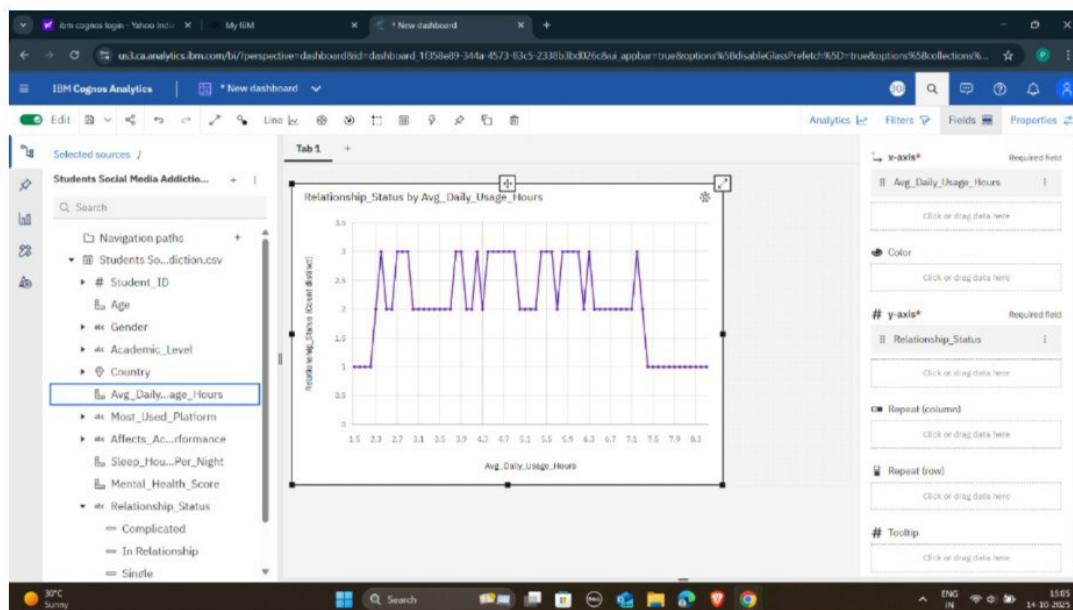
Step1:-Select Data: Load the Students_Social_Addiction.csv data file for analysis



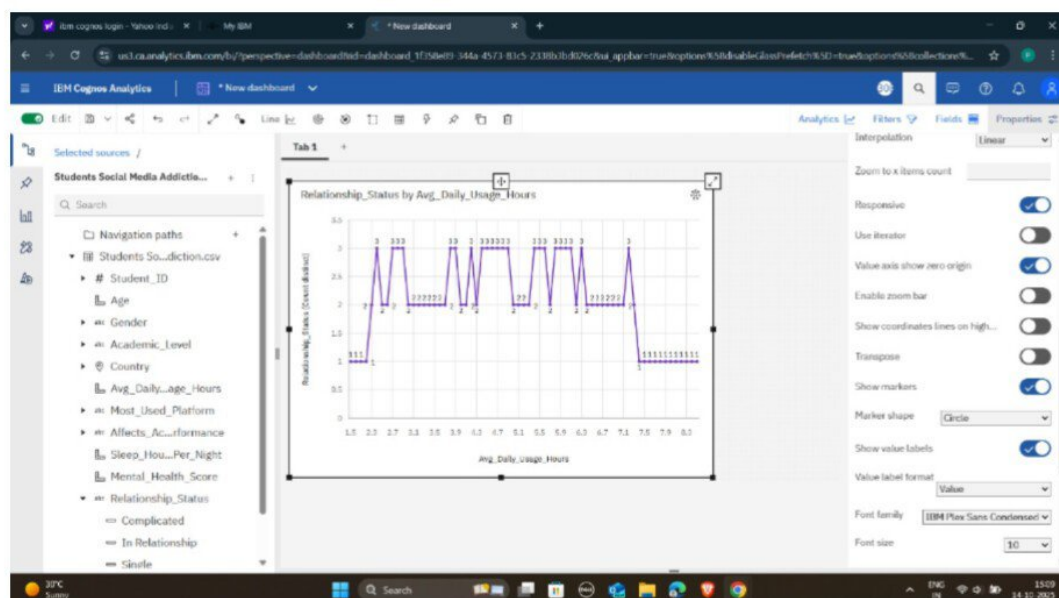
Step2:-Create Chart: Initialize a Line chart in a new dashboard.



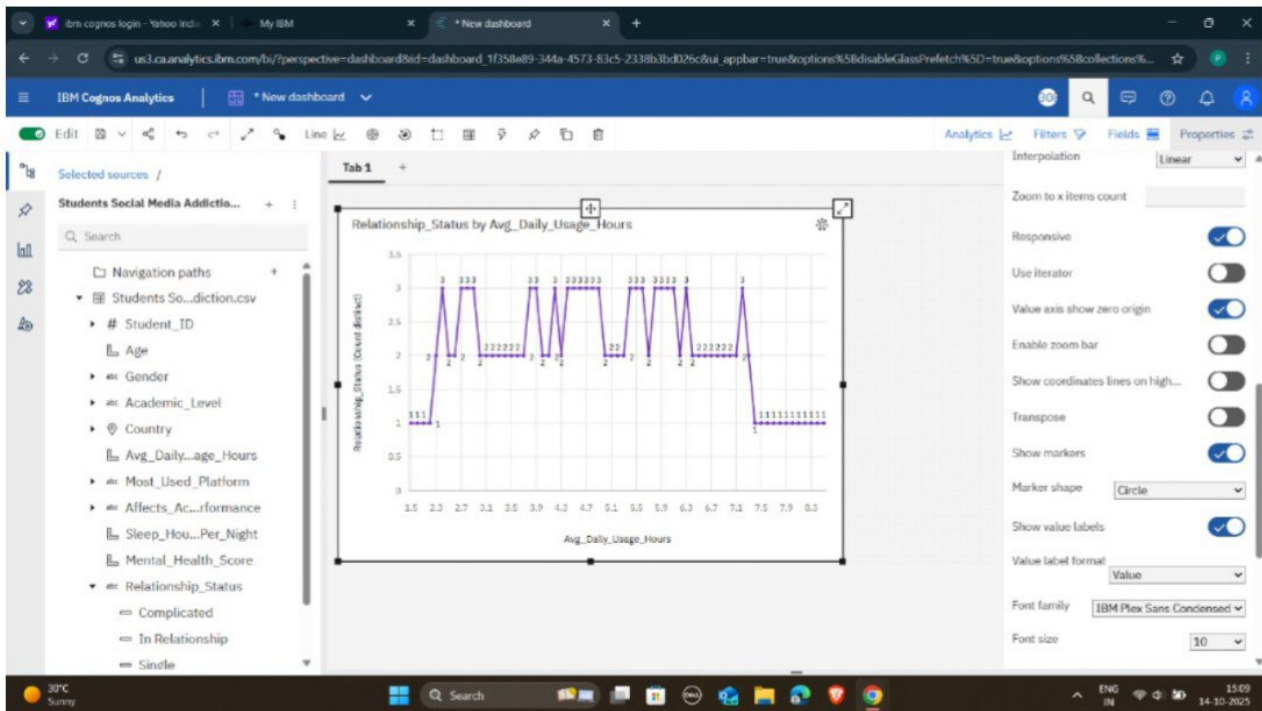
Step3:-Map Variables: Assign Avg_Daily_Usage_Hours to the X-axis and Relationship_Status (Count) to the Y-axis.



Step4:- Refine Visualization: Apply visualization properties such as showing value labels and setting the marker shape to 'Circle'.



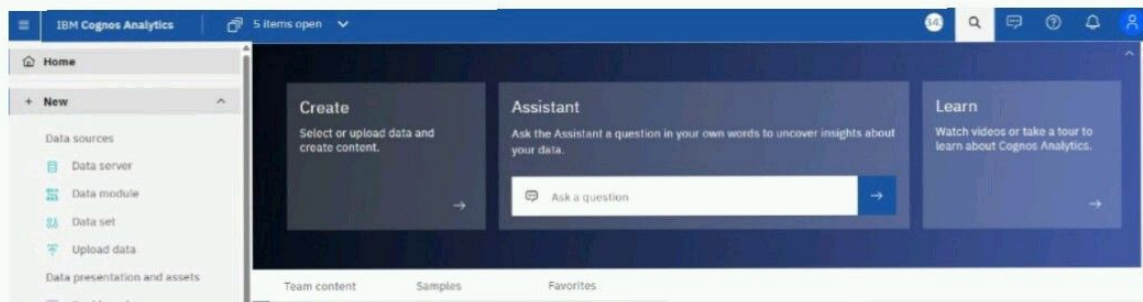
Save Dashboard: Save the final visualization with the title RELATIONSHIP VS ENGAGEMENT LEVEL.



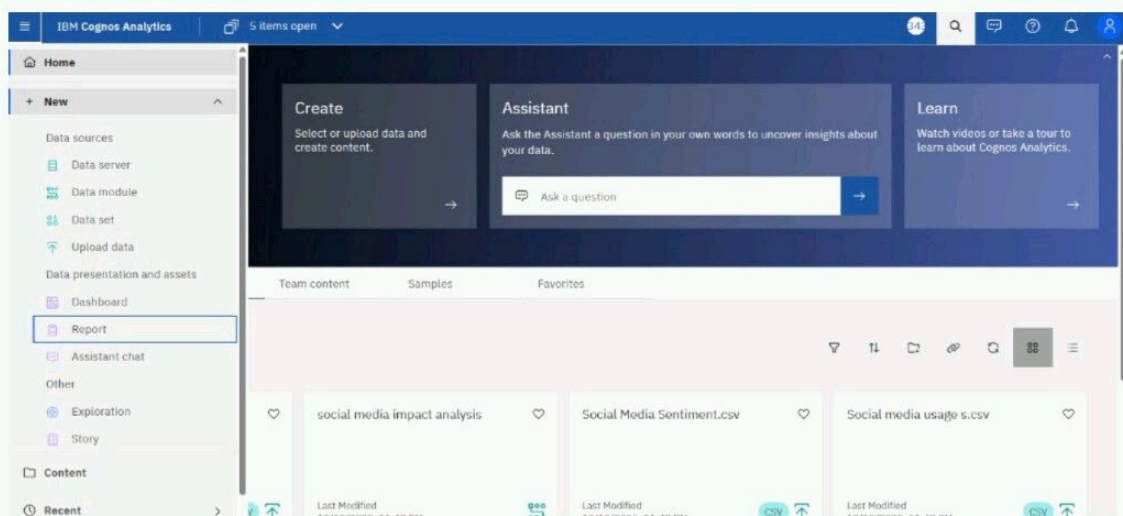
Project

Determine the effect of social media addiction on emotional well-being scores.

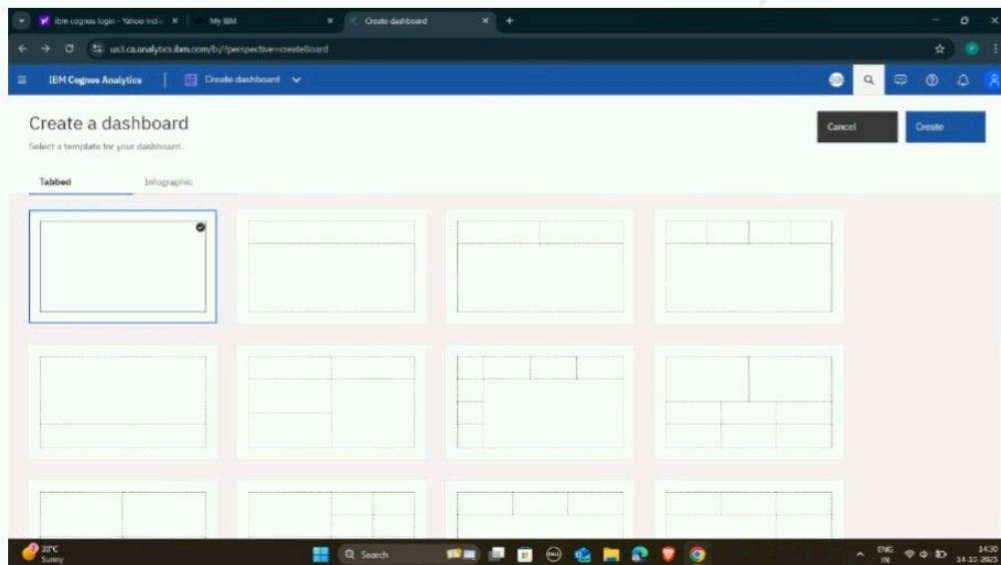
Step:1 - Logging to IBM and followed interface will be opened



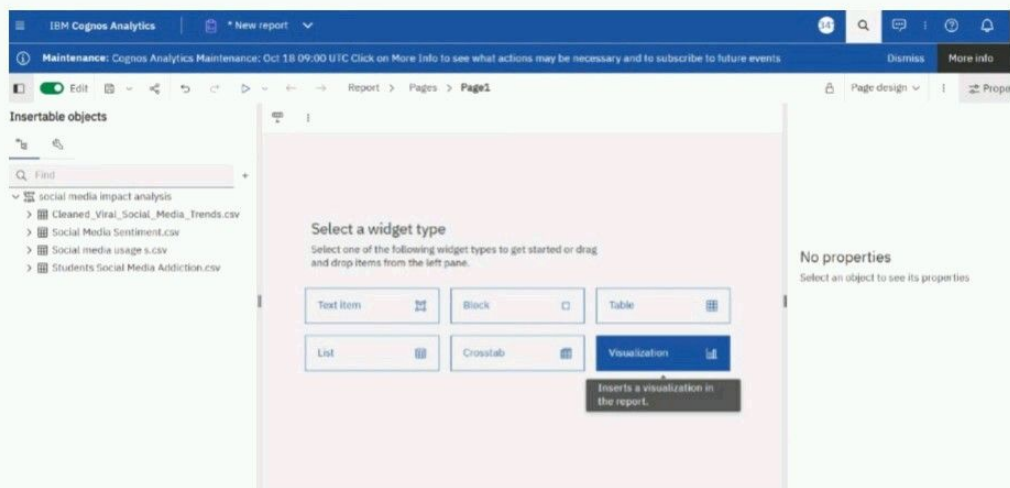
Step:2 - Click three line icon and select new -> Report option.



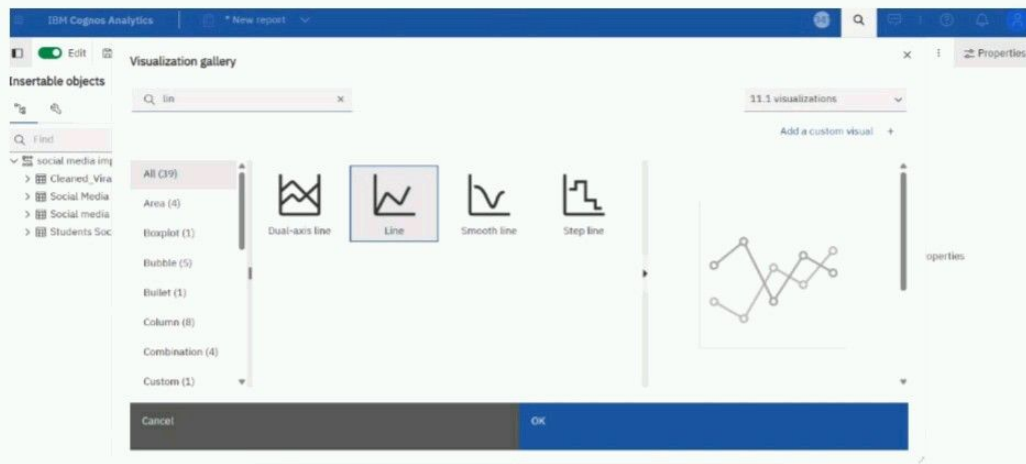
Step:3 - Click on Blank Report option and then click on create option.



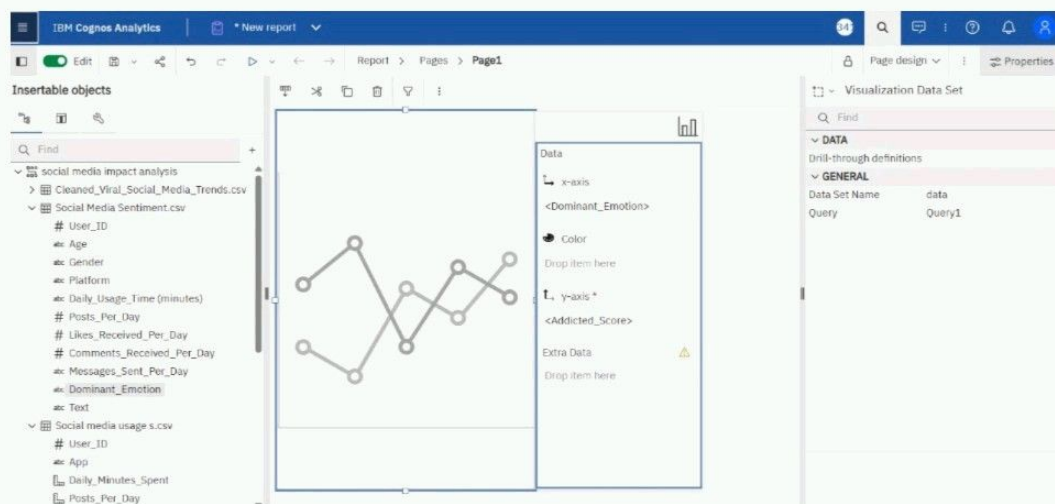
Step:4 - Select a visualization Report option.



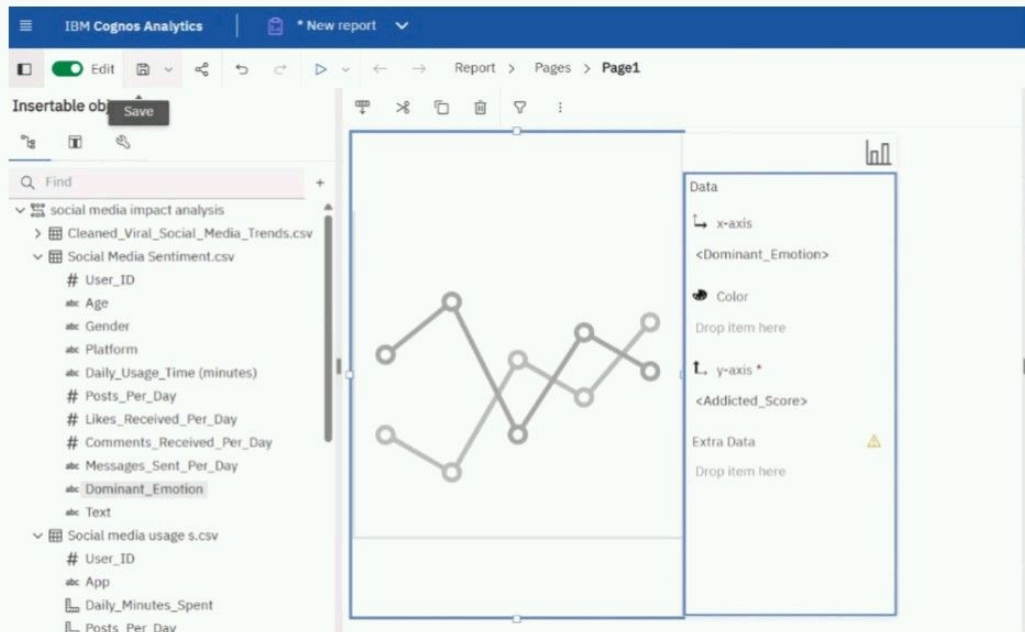
Step:5 - Select the Line chart from the Visualization gallery and click OK.



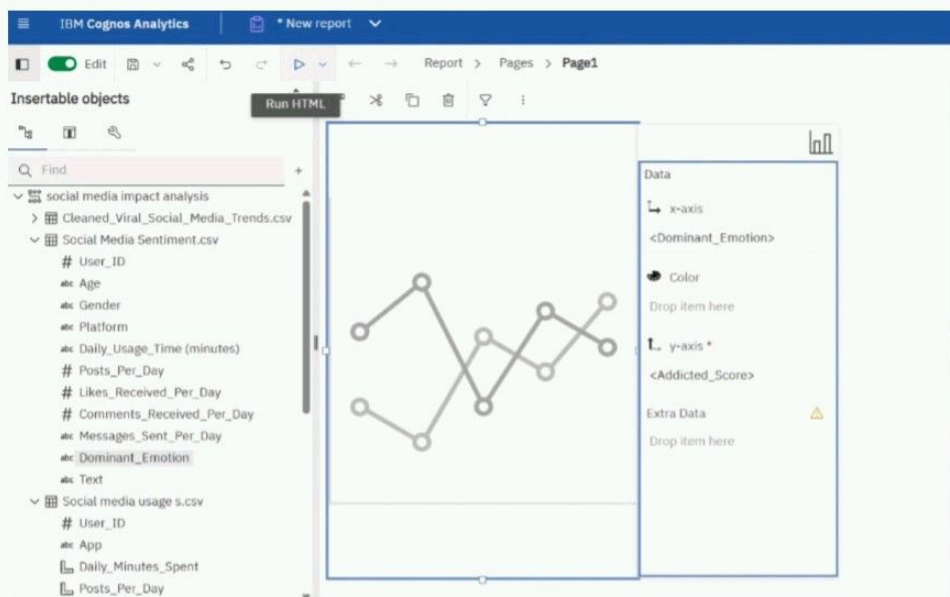
Step:6 - Drag <Dominant_Emotion> to the X-axis and <Addicted_Score> to the Y-axis* data slots.



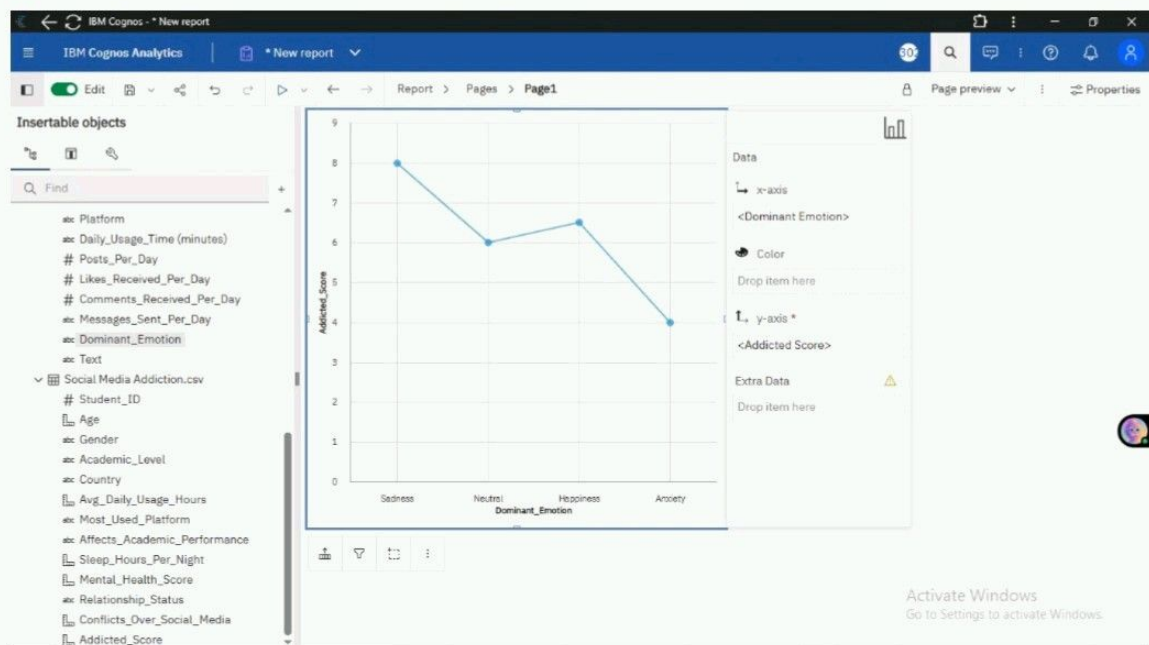
Step:7 - Click on the save to store the report.



Step:8 - Click on the Run to execute the Report.



Review the final line visualization showing the Addicted Score trend across different Dominant Emotions.

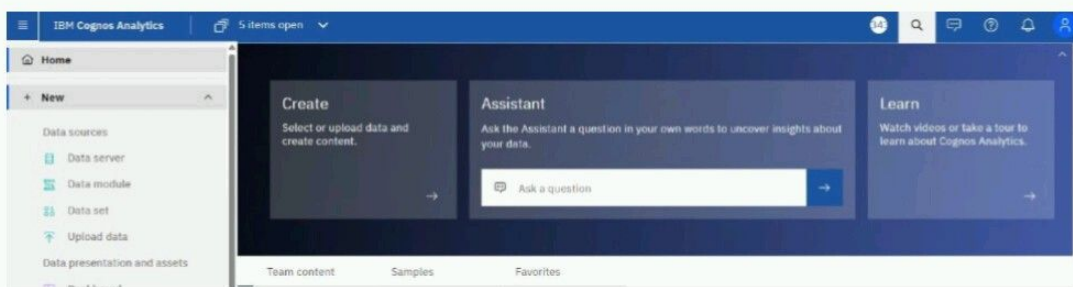


Project

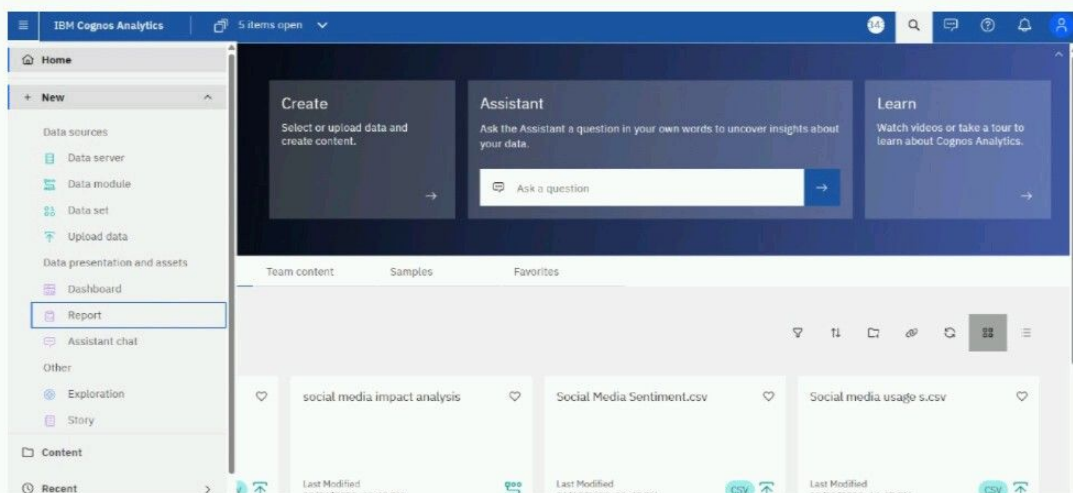
Identify the top 10 most engaging social media posts or trends

Solution: create a list using data module

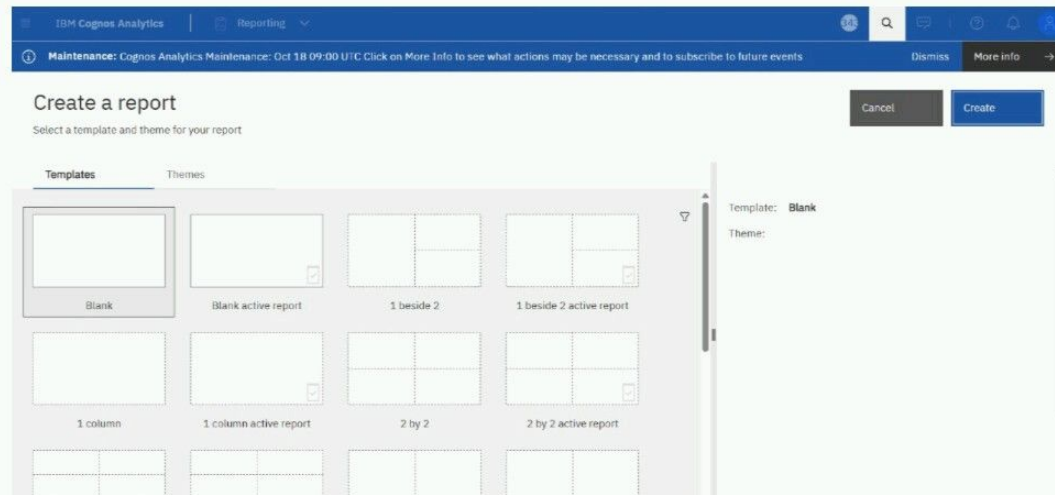
Step1 - Logging to IBM and followed interface will be opened



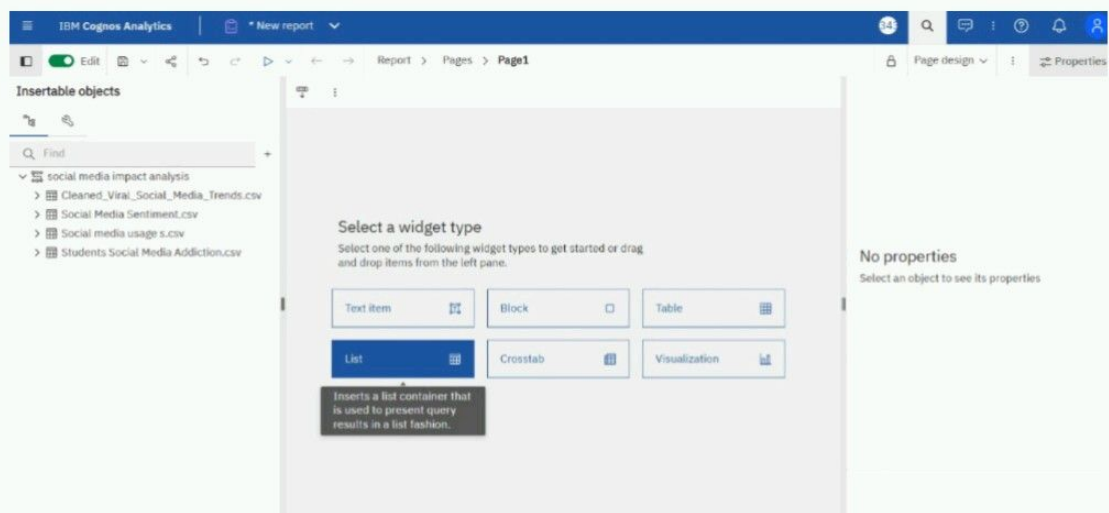
Step2 - Click three line icon and select new ->Report option.



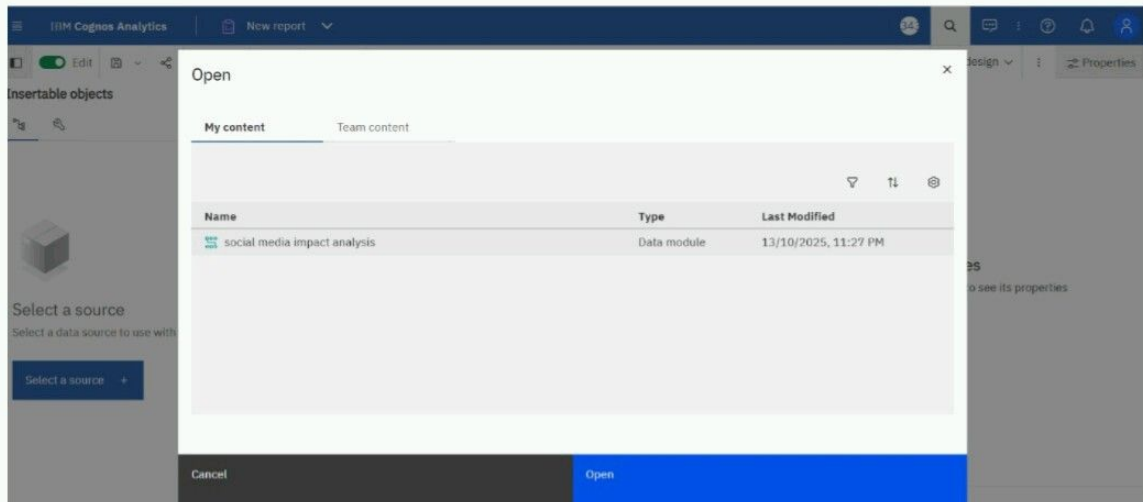
Step3 - Click on Blank Report option and then click on create option.



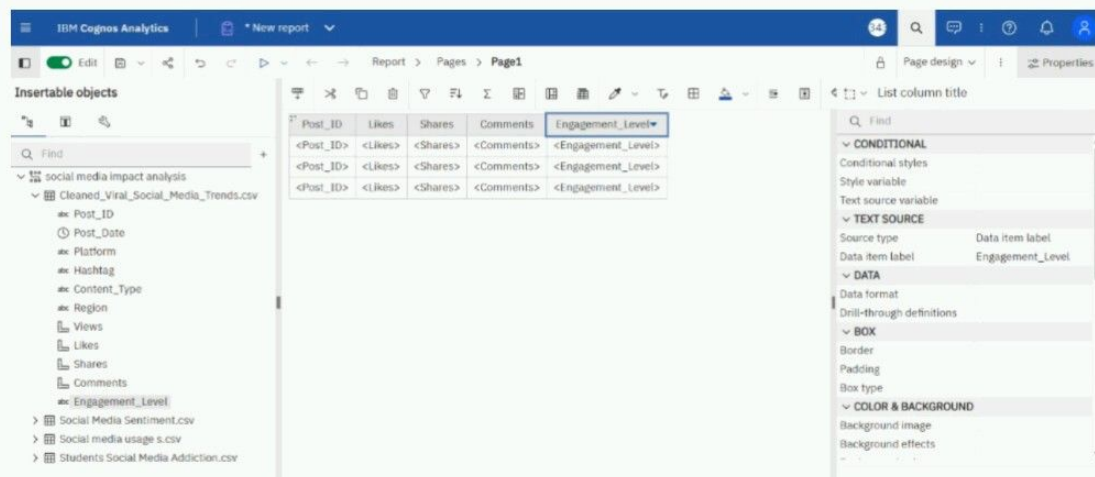
Step4 - Select a List Report Option



Step5 - Select Data module from my content option and click open.

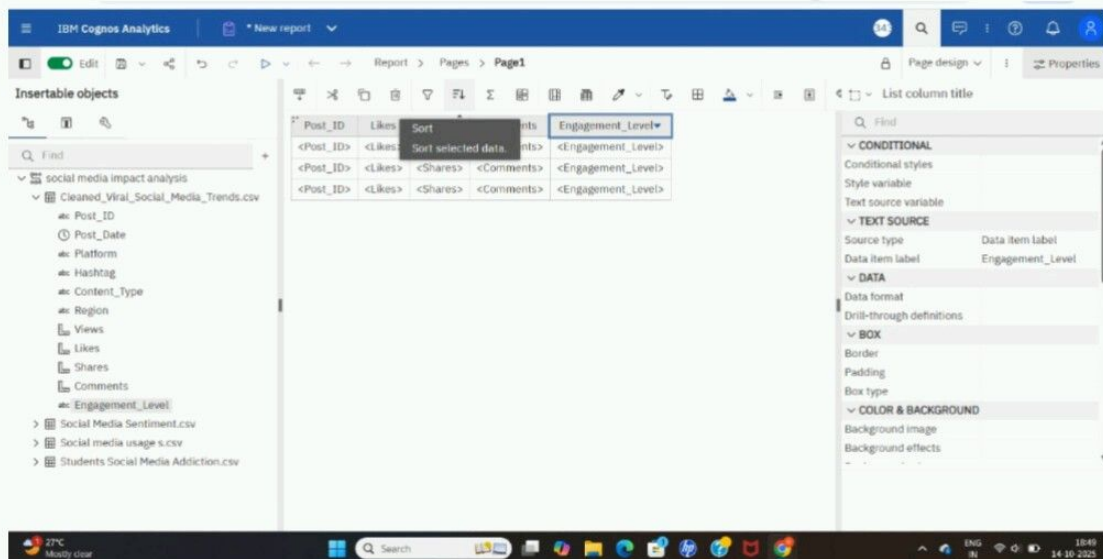


Step6 - Drag post_ID ,Likes, Share , Comments and Engagement _Level into the List Report.

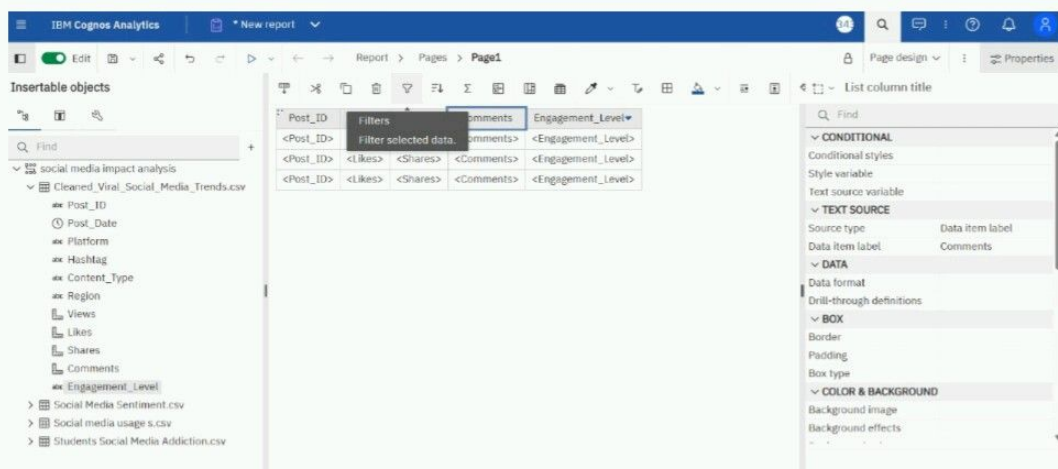


Step7 - Click on the column heading "Total Engagement"

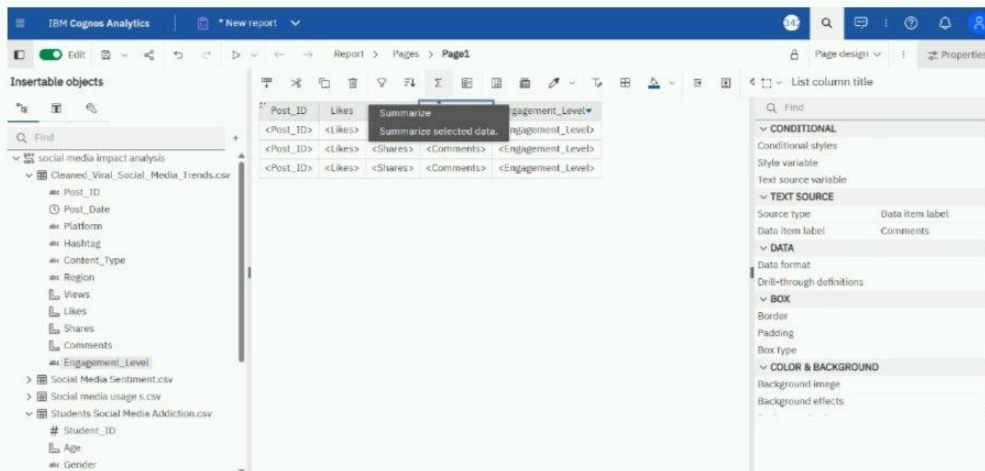
Choose "Sort → Descending" so that the highest engagement appears first.



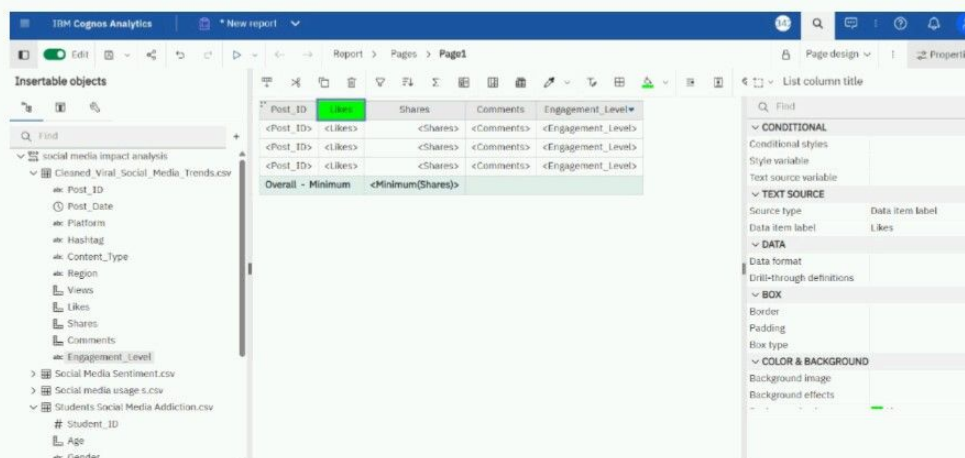
Step8 - Click on the Total Engagement column → choose "Filter" → "Top/Bottom".



Step:9 - Select the field 'Share Data' and apply Summarize → Minimum to display the lowest (minimum) value of the shared data.



Step:10 - Applied color formatting to highlight the Like Data visually.



Step:11 - Click -> Run.

IBM Cognos Analytics | New report

Insertable objects

Find

- social media impact analysis
 - Cleaned_Viral_Social_Media_Trends.csv
 - Post_ID
 - Post_Date
 - Platform
 - Hashtag
 - Content_Type
 - Region
 - Views
 - Likes
 - Shares
 - Comments
 - Engagement_Level
 - Social Media Sentiment.csv
 - Social media usage s.csv
 - Students Social Media Addiction.csv
 - Student_ID
 - Age
 - Gender

IBM Cognos Analytics | New report

Run HTML

Insertable objects

Find

- social media impact analysis
 - Cleaned_Viral_Social_Media_Trends.csv
 - Post_ID
 - Post_Date
 - Platform
 - Hashtag
 - Content_Type
 - Region
 - Views
 - Likes
 - Shares
 - Comments
 - Engagement_Level
 - Social Media Sentiment.csv
 - Social media usage s.csv
 - Students Social Media Addiction.csv
 - Student_ID
 - Age
 - Gender

Post_ID	Likes	Shares
<Post_ID>	<Likes>	<Shares>
<Post_ID>	<Likes>	<Shares>
<Post_ID>	<Likes>	<Shares>
Overall - Minimum	<Minimum(Shares)>	

You'll see table like this ->

IBM Cognos Analytics | New report

Maintenance: Cognos Analytics Maintenance: Oct 18 09:00 UTC

HTML

Post_ID	Likes	Shares	Comments	Engagement_Level
Post_2	2,15,240	65,860	27,239	Medium
Post_5	1,71,361	69,581	6,376	Medium
Post_28	3,22,601	35,671	48,768	Medium
Post_70	4,21,149	53,769	27,804	Medium
Post_71	50,577	92,342	43,775	Medium
Post_78	1,32,163	29,340	40,557	Medium
Post_81	2,11,176	66,264	14,337	Medium
Post_95	4,44,670	64,311	7,956	Medium
Post_110	1,88,191	4,660	26,605	Medium
Post_127	2,77,223	81,798	46,985	Medium
Post_128	1,40,163	85,932	40,674	Medium
Post_161	56,569	59,897	49,341	Medium
Post_168	3,82,792	78,654	30,532	Medium
Post_176	36,274	51,278	44,234	Medium
Post_190	3,37,321	99,486	26,524	Medium
Post_199	1,31,915	64,092	22,170	Medium

Top Page up Page down Bottom